

MONEY, MASTERED · THE ROADMAP

BOOK FIVE OF EIGHT

CANDLESTICKS · TRENDS · CHART PATTERNS · INDICATORS · RISK · HONESTY



READING THE PRICE CHART — HONESTLY

Technical Analysis

The other half of the picture: not the business, but the price itself. Candlesticks, trends, patterns and indicators — what each really means, the actual math behind them, and a frank account of what charts can and cannot do.

REAL CHARTS · REAL FORMULAS · FIVE PARTS · EIGHTEEN CHAPTERS · BUILDS
ON BOOKS 1–4

Technical Analysis. Book Five of *Money, Mastered — The Roadmap*. First Edition · 2026. It builds on Books 1–4 and assumes you know what a share, a market and a price chart are.

A note on the charts. The price charts in this book are *drawn* to illustrate each concept clearly — they are realistic teaching diagrams, not snapshots of any specific stock on any specific day. In them, an **up** candle (close above open) is shown dark and a **down** candle gold; your trading app will typically use green for up and red for down. Currency is shown as \$ as a stand-in for any currency.

Education, not advice — and an unusually important warning here. Technical analysis is widely *oversold*. This book teaches it properly *and* tells you the truth: it is a tool of probability, not prophecy, and the data on short-term trading is grim. Nothing here is investment advice, a trading system, or a promise of profit. No pattern or indicator predicts the future reliably.

Set in Fraunces, Spectral & Archivo. All charts & diagrams are original vector graphics; the book is self-contained and works offline.

READ THIS BEFORE YOU TRADE ON ANYTHING IN THIS BOOK

The large majority of active short-term traders lose money after costs — this is one of the most consistent findings in all of finance, and regulators worldwide repeat it. Technical analysis can sharpen *when* you act on a decision you already have good reason to make; it cannot turn chart-reading into a money machine. If anyone sells you “sure-shot” signals or a “guaranteed” system, they are taking your money, not making you rich. Treat this book as a way to understand price behaviour — and to protect yourself — not as a licence to gamble.

THE ROADMAP

Where This Book Sits

Book 4 judged the business. This one studies the price chart. Together they give you both lenses on a stock — what it's worth, and what its price is doing.

STAGE 1 · FOUNDATIONS

Book 1 · Money, From Zero DONE

Book 2 · Where to Put Your Money DONE

STAGE 2 · THE MARKETS

Book 3 · How the Market Works DONE

Book 4 · Fundamental Analysis DONE

Book 5 · Technical Analysis **YOU ARE HERE**

Book 6 · Funds, SIPs & Portfolios PUTTING IT TOGETHER

Book 7 · Derivatives, Tax, Psychology & Safety ADVANCED & PROTECTION

STAGE 3 · WEALTH

Book 8 · Building & Protecting Wealth THE LONG GAME

A caution before you start: most long-term investors (Books 1–2) can succeed without ever using technical analysis. This book is for understanding price behaviour and for those who choose to time entries carefully — always on top of, never instead of, sound judgement.

CONTENTS · BOOK FIVE

What's Inside

Eighteen chapters in five parts. We build from how to read a chart, through candlesticks (with real charts of every major pattern) and chart patterns, to indicators (with the actual math), and end with risk management and an unusually honest reckoning.

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WELCOME

How to Use This Book

Technical analysis is the most over-promised subject in all of investing — and one of the most misunderstood. This book does something rare: it teaches the craft properly, with real charts and real math, while telling you the honest truth about its limits.

If Book 4 asked “what is this business worth?”, this book asks a completely different question: “what is the price actually *doing*?” Technical analysis studies the chart — the footprints left by every buyer and seller — to read the mood of the market and to time decisions. Used with discipline and humility, it's a genuine skill. Sold as a crystal ball, it's a trap that has emptied countless accounts.

We'll build carefully. **Part I** teaches you to read any chart — trend, support and resistance, the grammar of price. **Part II** is the heart: candlesticks, with a real drawn chart of every major pattern. **Part III** covers the larger chart patterns. **Part IV** tackles indicators — moving averages, RSI, MACD, Bollinger Bands — and, unlike most books, actually shows you *the formulas and how to compute them*. **Part V** is where the real money is made or saved: risk management, and a brutally honest chapter on what trading is really like.

How each chapter is built

Every chapter opens with **what you'll learn**, unfolds in numbered **sub-chapters** with **real charts**, and closes with **Key Takeaways** and a **Self-Check**. Indicator chapters include **Worked Calculations** with the actual numbers. Watch for **In Plain Terms** explainers and frequent **Watch Out** warnings — there are more here than in any other book in the series, by design.

A chart shows you what has happened and hints at the mood of the crowd. It does not show you the future. Hold both halves of that sentence at once and you'll use this book well.

One last framing. Nothing in this book contradicts Books 1–4: the surest path to wealth remains owning good assets cheaply and patiently. Technical analysis is a *supplement* — a way to understand price behaviour and, for those who choose to, to time entries more carefully. It is never a substitute for sound investment judgement, and it is never a shortcut.



Let's begin honestly — with what technical analysis really is, and what it can and cannot do.

I

PART ONE

The Foundations

Before any pattern or indicator, you need the grammar of price: what a chart is, what a trend is, and where prices tend to pause. Master these four ideas and most of technical analysis becomes common sense.

WHAT TA IS

| READING A CHART

| TREND

| SUPPORT & RESISTANCE

CHAPTER ONE

What Technical Analysis Is (and Isn't)

Technical analysis is the study of price itself — the chart, the volume, the rhythm of buying and selling — to gauge the mood of the market. Done honestly, it's a real skill. Sold as fortune-telling, it's the most expensive fantasy in finance. We'll be clear about which is which from page one.

IN THIS CHAPTER YOU WILL LEARN

- The core idea behind technical analysis.
- How it differs from fundamental analysis (Book 4).
- The honest truth: what charts can and can't do.
- The three assumptions technical analysis rests on.

Two investors look at the same stock. One (the fundamental analyst of Book 4) studies the business — its profits, debts and moat. The other studies only the *chart* — how the price has moved and on what volume — caring little about what the company even makes. That second investor is the technical analyst, and this book is about their craft: reading the footprints that buyers and sellers leave in the price.

1.1 The Core Idea

Technical analysis rests on a simple premise: **the price chart already reflects everything** — every fact, hope, fear and rumour that anyone acting on the stock currently holds. Rather than trying to dig out new information about the business, the technician studies the *result* of everyone's collective decisions: the price. And because markets are made of humans, and human emotions (greed, fear, hope) repeat, the patterns those emotions carve into a chart tend to *recur*. Technical analysis is, at heart, the study of crowd psychology written in the language of price and volume.

1.2 Technical vs. Fundamental

COMPARE & UNDERSTAND · TWO DIFFERENT QUESTIONS

	FUNDAMENTAL (BOOK 4)	TECHNICAL (THIS BOOK)
Studies	The business — profits, assets, moat	The price chart — trend, patterns, volume
Asks	“What is it worth?”	“What is the price doing?”
Time horizon	Usually years	Often days to months
Best for	Deciding <i>what</i> to own	Helping decide <i>when</i> to act
Core risk	Being wrong about value	Reading meaning into randomness

They aren't enemies — many investors use fundamentals to choose *what* to buy and a little technical sense to time *when*. But never confuse them: a beautiful chart pattern on a terrible business is still a terrible business, and the surest long-term wealth (Books 1–2) comes from value and patience, not chart-reading.

1.3 The Honest Truth — What Charts Can and Can't Do

Here is the candour this subject badly needs. Charts **can**: show you the trend and the mood, mark sensible levels to act or to cut losses, impose discipline, and occasionally flag when a crowd is getting euphoric or panicked. Charts **cannot**: predict the future reliably, turn a coin-flip into a sure thing, or overcome costs and human emotion to make most active traders rich — because they don't. A great deal of what looks like a “pattern” is, statistically, noise the pattern-hungry brain has connected into a story. Use technical analysis as a *probabilistic aid and a discipline tool*, never as prophecy. The honest technician speaks in odds and risk, never in certainty.

In Plain Terms · the self-fulfilling part

Some technical levels “work” partly because *so many people watch them*. If thousands of traders believe a stock will bounce at \$100 and all place buy orders there, their combined buying can actually *cause* the bounce — for a while. This self-fulfilling element is real and useful, but it's also fragile: when the crowd's belief breaks, the level breaks hard.

Technical analysis works best where it reflects genuine human behaviour, and fails where it's just superstition dressed up in jargon.

1.4 The Three Assumptions

Classical technical analysis rests on three foundational beliefs, worth stating plainly so you can judge them yourself. **One: price discounts everything** — all known information is already in the price, so you can focus on the price alone. **Two: prices move in trends** — once a direction is established, it's more likely to continue than to reverse, until clear evidence says otherwise. **Three: history tends to repeat** — because human psychology is constant, the patterns of the past recur in the future. None of these is an iron law; the first is debatable and the third is only a tendency. But together they explain *why* technicians do what they do — and holding them loosely, as tendencies rather than certainties, is the mark of a sensible practitioner.

WATCH OUT

The single biggest danger in this entire book

It is this: **technical analysis makes you feel in control of the uncontrollable.**

Staring at a chart, you sense you can see the future — and that illusion fuels overtrading, overconfidence, and the steady losses that follow. The data (Chapter 18) is brutal: most active traders lose. As you learn these tools, repeat to yourself that you are reading *probabilities and crowd mood*, not destiny. The moment a chart makes you feel certain is the moment to be most careful.

KEY TAKEAWAYS

1. **Technical analysis studies the price chart** — the footprints of crowd psychology — not the business.
2. **It asks “what is the price doing?”**; fundamental analysis asks “what is it worth?” Use them for different jobs.
3. **Charts aid probability and discipline; they do not predict the future** — much apparent “pattern” is noise.
4. **Three assumptions:** price discounts everything, prices trend, history rhymes — tendencies, not laws.

SELF-CHECK

1. In one sentence, what does technical analysis study, and why?
2. How does the technical question differ from the fundamental one?
3. Give one thing charts can do and one thing they cannot.
4. State the three assumptions, and say why each is a tendency rather than a law.

CHAPTER TWO

Reading a Price Chart

Before you can spot a pattern you must read the chart itself — its types, its time frames, and the volume bars beneath it. This is the alphabet of technical analysis; everything else is spelling.

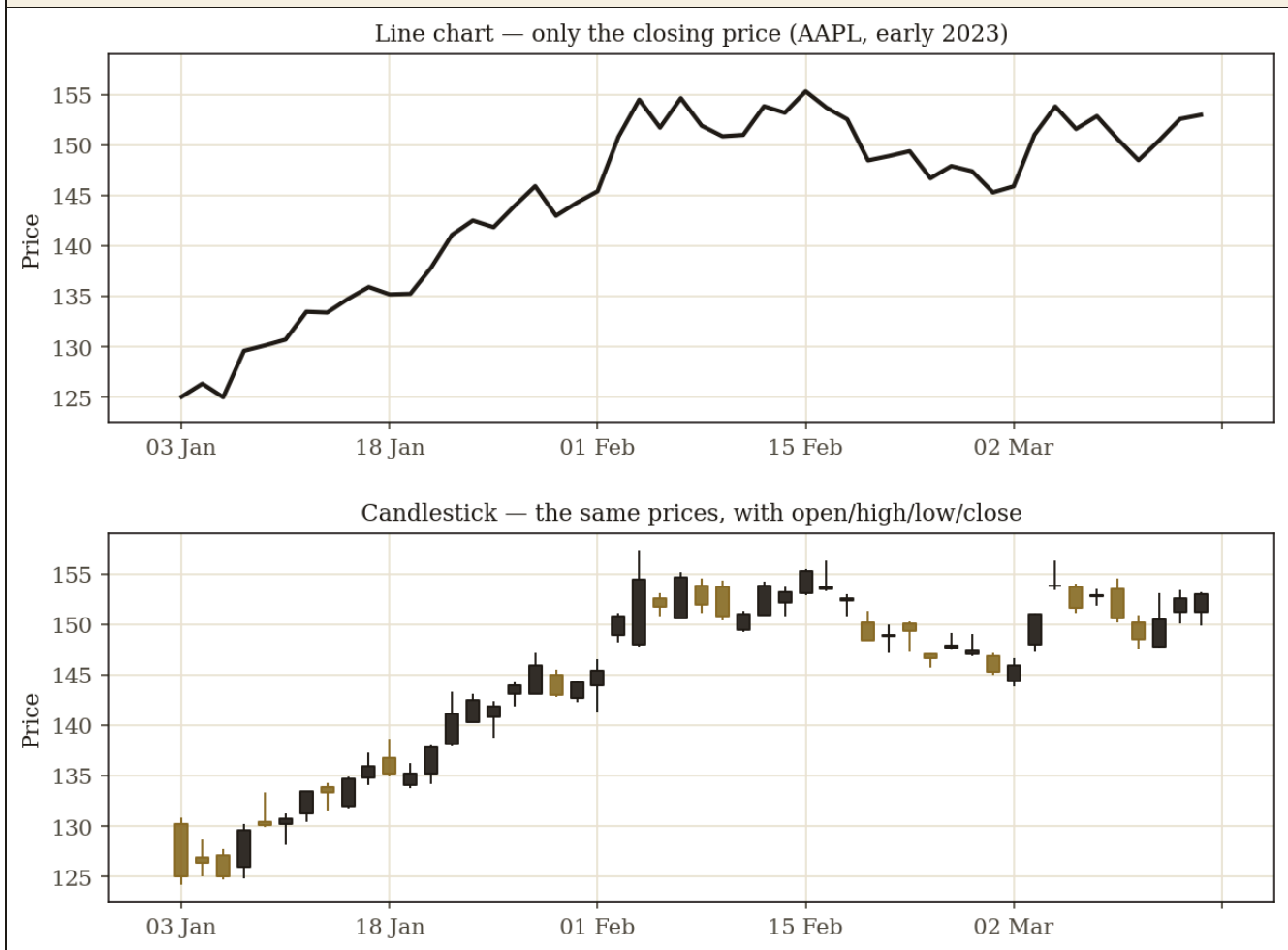
IN THIS CHAPTER YOU WILL LEARN

- The three main chart types — line, bar and candlestick.
 - Why the time frame you choose changes everything.
 - What volume is, and why it confirms (or doubts) a move.
-

A price chart plots one thing — price (vertical) against time (horizontal) — but *how* it plots it, and over what period, dramatically changes what you see. Let's learn to read it before we read into it.

2.1 Line, Bar & Candlestick

Three ways to draw the same prices. A **line chart** simply connects each period's closing price — clean and good for seeing the big trend. A **bar chart** shows four prices per period (open, high, low, close) as a vertical bar with ticks. A **candlestick chart** shows those same four prices but as a far more readable “candle” (Part II) — and it has become the technician's default because the eye reads it instantly. Here is the same week of prices drawn as a line and as candlesticks:

FIGURE 2.1 · THE SAME PRICES, TWO WAYS — APPLE, EARLY 2023

Real data, both ways. *The same Apple daily prices shown as a line (closing prices only, top) and as candlesticks (open/high/low/close, bottom). The candles reveal the intra-day battle the line hides — which is why technicians work in candlestick charts.*

2.2 Time Frames Change Everything

One chart can tell two opposite stories depending on the **time frame** — the period each candle represents. On a 5-minute chart a stock might look like it's crashing; on a weekly chart that same drop is an invisible blip in a steady uptrend. Neither is “wrong”; they answer different questions. Short time frames (minutes, hours) suit short-term traders and are dominated by noise; long time frames (days, weeks, months) suit investors and reveal the durable trend. The golden rule: **match your time frame to your intentions**, and beware drawing long-term conclusions from a short-term chart (or vice versa). Always know which time frame you're looking at before you react to it.

2.3 Volume — the Conviction Behind the Move

Beneath almost every price chart sits a row of bars showing **volume** — how many shares traded in each period. Volume is the technician's lie-detector, because it measures *conviction*. A price rise on *heavy* volume means many participants are committing real money — a believable, well-supported move. The same rise on *thin* volume means few are participating — a flimsy move that can reverse easily. The principle to remember: **volume should confirm the move**. A breakout or a trend backed by rising volume is far more trustworthy than one on a trickle. When price does one thing and volume disagrees, doubt the price.

WATCH OUT

Don't react to the wrong time frame

A huge share of beginner panic comes from staring at short time frames. A long-term investor checking a 1-minute chart will see terrifying squiggles that mean nothing for their decade-long plan — and may sell in fear over noise. If you're investing for years, look at weekly or monthly charts and ignore the intraday chaos. The time frame you watch quietly shapes your emotions and decisions; choose it deliberately.

KEY TAKEAWAYS

1. **Three chart types:** line (closes), bar, and candlestick (open/high/low/close) — candlesticks are the technician's default.
2. **Time frame changes the story;** match it to your intentions and don't mix horizons.
3. **Volume measures conviction** — it should confirm the move; price up on thin volume is suspect.
4. **Watch the right time frame for your goal** to avoid reacting to noise.

SELF-CHECK

1. What extra information does a candlestick show that a line chart doesn't?
2. Why can the same stock look bullish and bearish at once?

3. What does volume measure, and what should it do relative to a price move?
4. If you invest for ten years, which time frame should you mostly watch — and why?

CHAPTER THREE

Trend — the Most Important Concept

If you learn only one thing from technical analysis, learn to identify the trend. Almost every technical idea — patterns, indicators, signals — is ultimately a way of answering one question: which way is this market really moving?

IN THIS CHAPTER YOU WILL LEARN

- The three directions a market can move.
- How to define a trend objectively (highs and lows).
- How to draw and use a trendline.
- Why “the trend is your friend” — with a vital caveat.

Markets don't move in straight lines; they move in waves that, taken together, lean in a direction. That direction is the **trend**, and identifying it correctly is the foundation everything else is built on. Trade with the trend and the odds tilt your way; fight it and you're swimming against the current.

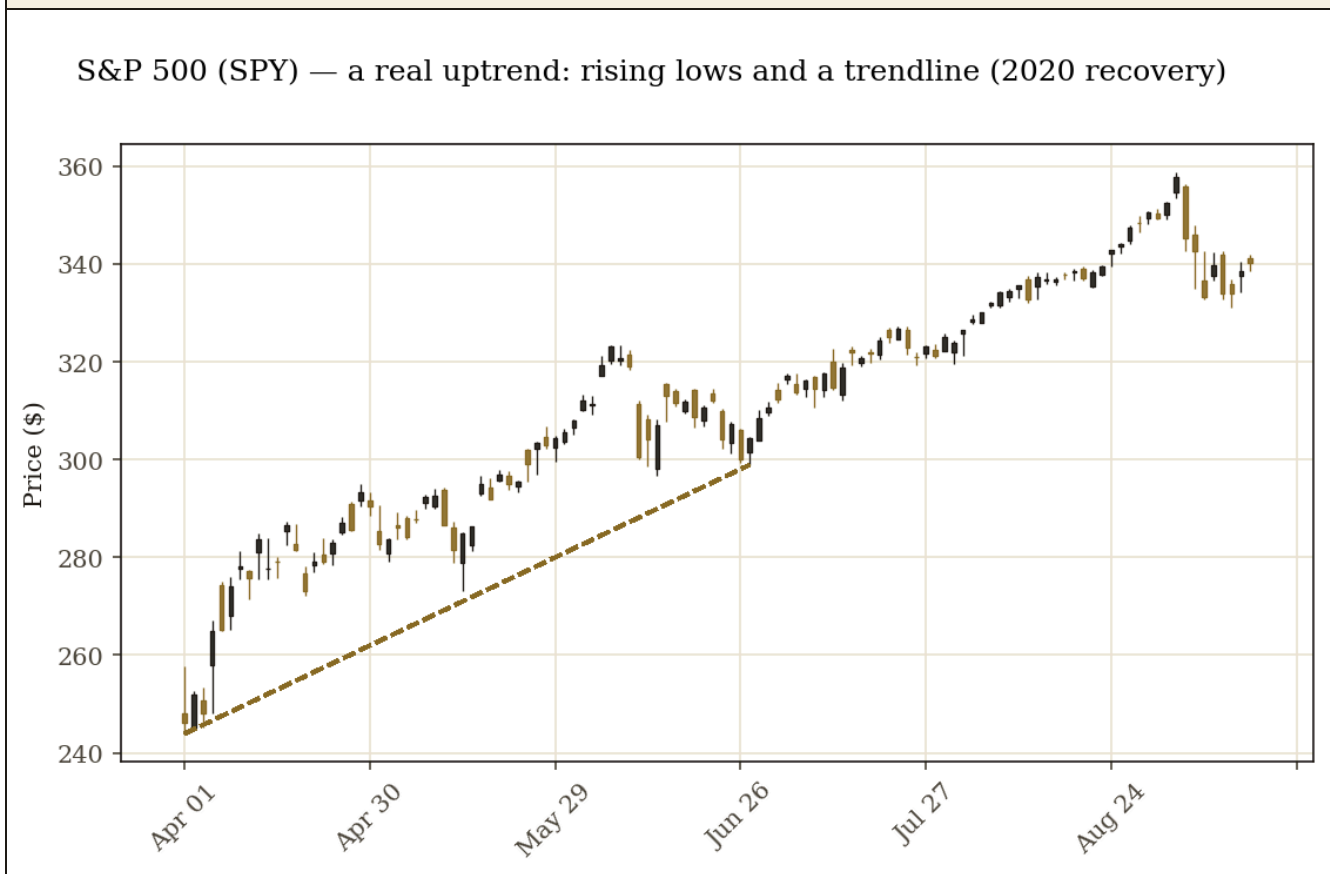
3.1 Up, Down, or Sideways

At any time, a market is doing one of three things: trending **up** (rising over time), trending **down** (falling), or moving **sideways** (ranging, with no clear direction). It sounds obvious, but most trading errors come from misjudging this — buying as if in an uptrend when the market has actually rolled over, or selling in fear during a healthy uptrend's normal dip. Sideways (or “ranging”) markets are their own challenge: trend-following tools work poorly, and many traders simply lose money churning in a market that's going nowhere.

3.2 Defining a Trend Objectively

Feelings aren't enough; a trend has an objective definition. An **uptrend** is a series of *higher highs and higher lows* — each peak and each dip is higher than the last. A **downtrend** is *lower highs and lower lows*. The moment that pattern breaks — when an uptrend prints a *lower low*, say — the trend may be turning. This simple rule turns a vague impression into a checkable fact.

FIGURE 3.1 · A REAL UPTREND AND ITS TRENDLINE — S&P 500, 2020



The staircase up. Real SPY data through the 2020 recovery: each pullback bottoms higher than the last, and a trendline drawn under the rising lows supports the advance. While price holds above that line, the uptrend is intact.

3.3 Trendlines

A **trendline** makes the trend visible: in an uptrend, draw a straight line connecting the rising lows; in a downtrend, connect the falling highs. The line acts as a moving floor (or ceiling) that price respects again

and again — bouncing off it during the trend. Trendlines do two useful jobs: they confirm the trend is intact while price holds the line, and they flag a possible reversal when price decisively breaks through it. They're among the simplest and most genuinely useful tools in technical analysis — though, like everything here, they break sometimes, and a break needs confirmation (volume, a follow-through) before you trust it.

3.4 “The Trend Is Your Friend” — With a Caveat

The oldest maxim in technical analysis is “*the trend is your friend*.” Its wisdom: trends tend to persist, so trading *with* the prevailing direction puts probability on your side, while trying to pick tops and bottoms (fighting the trend) is a fast way to lose. But the full saying has a crucial ending often left off: “... *until the end, when it bends*.” Trends do reverse, and the trader who rides one too stubbornly gives back their gains at the turn. So: respect the trend, trade with it — but watch for the higher-high/higher-low pattern breaking, which is your signal that the friend is leaving.

WATCH OUT

Don't try to catch the falling knife

Beginners love to “buy the dip” in a downtrend, certain the bottom is in — and get cut again and again as it keeps falling (lower lows). Catching a falling knife — trying to call the exact bottom of a downtrend — is one of the most reliable ways to lose money. The trend-following discipline says: wait for the downtrend to actually *stop* making lower lows and show signs of turning *before* buying. Patience beats heroics.

KEY TAKEAWAYS

1. **Identifying the trend is the foundation** of technical analysis — up, down, or sideways.
2. **Define it objectively:** uptrend = higher highs & higher lows; downtrend = lower highs & lower lows.
3. **Trendlines** connect the rising lows (or falling highs) and confirm the trend until decisively broken.
4. **Trade with the trend** — but watch for the pattern breaking, and never catch a falling knife.

SELF-CHECK

1. What are the three things a market can be doing?
2. Define an uptrend and a downtrend using highs and lows.
3. How do you draw a trendline in an uptrend, and what does a break of it suggest?
4. What's the full version of “the trend is your friend,” and why does the ending matter?

CHAPTER FOUR

Support & Resistance

Prices don't move freely; they tend to pause, bounce and stall at certain levels — floors where buyers keep appearing, and ceilings where sellers keep emerging. These levels, called support and resistance, are the most useful map a technician can draw.

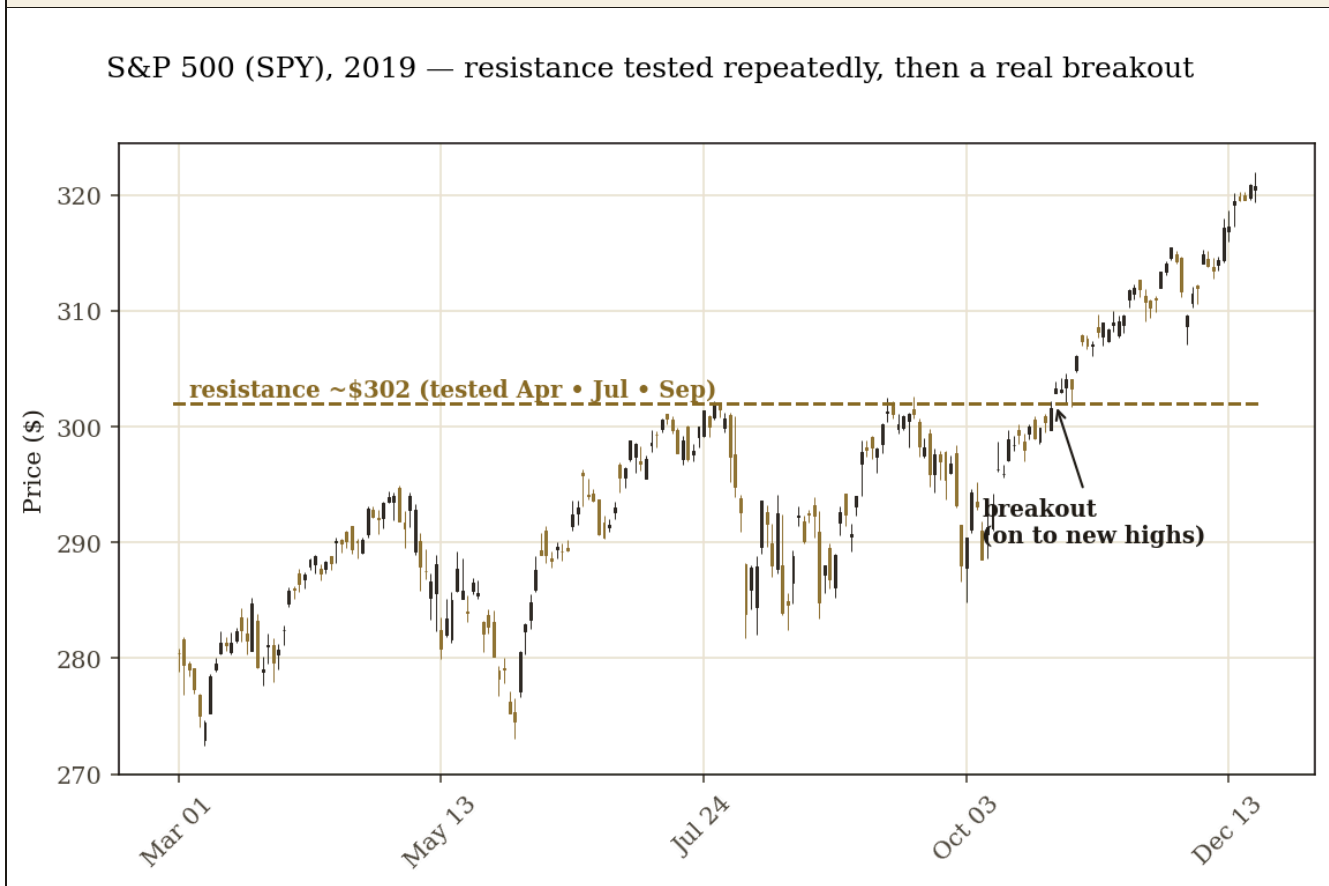
IN THIS CHAPTER YOU WILL LEARN

- What support and resistance are.
 - Why they form — the psychology beneath them.
 - What a breakout means, and how levels flip roles.
-

If trend tells you the direction, support and resistance tell you the *landmarks* along the way — the prices where something tends to happen. They're where technicians look to buy, to sell, and to place the stop-losses that protect them.

4.1 Floors and Ceilings

Support is a price level where falling tends to stop — a “floor” where buyers have repeatedly stepped in, their demand halting the decline and pushing price back up. **Resistance** is the opposite — a “ceiling” where rising tends to stall, as sellers repeatedly emerge and supply overwhelms demand. Price often oscillates between a support below and a resistance above, like a ball bouncing in a room.

FIGURE 4.1 · RESISTANCE TESTED, THEN BROKEN — S&P 500, 2019

The ceiling, and the escape. Real SPY data: the \$302 level capped the market repeatedly through mid-2019 (resistance), until a decisive breakout in late October carried price on to new highs. A breakout on real data — verify the dates yourself.

4.2 Why They Form — the Psychology

Support and resistance aren't magic lines; they're **memory** — the residue of past decisions and emotions. Resistance forms at a level where many people previously bought and then watched the price fall; when it climbs back to their entry price, they sell to “get out even,” creating a wall of supply. Support forms where buyers previously found value and remember it, stepping back in. Round numbers (\$100, \$50) also act as psychological levels simply because humans fixate on them. In short, these levels work to the extent that they reflect real, repeating human behaviour — which is exactly why they sometimes hold beautifully and other times shatter.

4.3 Breakouts and Role Reversal

When price **breaks** decisively through a level, two things matter. First, the break should ideally come on *rising volume* (Chapter 2) to be believable — a breakout on thin volume is often a “false breakout” that quickly reverses, trapping eager traders. Second, broken levels tend to **switch roles**: once price breaks above a resistance, that old ceiling often becomes a new floor (support) on any pullback — and a broken support often becomes new resistance. This “role reversal” is one of the most reliable behaviours in technical analysis, again because it mirrors how the crowd’s memory works: yesterday’s sellers become today’s buyers once they’re proven wrong.

WATCH OUT

False breakouts are everywhere

Markets are full of moves that *look* like a breakout, suck in eager traders, then snap back — sometimes deliberately, as larger players “hunt” the stop-losses clustered just beyond obvious levels. The defences: wait for *confirmation* (a close beyond the level, not just a brief poke; rising volume; a follow-through day) rather than jumping the instant price crosses a line. And never forget the broader lesson of Chapter 1 — even a confirmed breakout is a probability, not a promise.

KEY TAKEAWAYS

1. **Support is a floor (buyers step in); resistance is a ceiling (sellers appear).** Price oscillates between them.
2. **They form from memory and psychology** — past decisions and round numbers — not magic.
3. **Breakouts** through a level (ideally on rising volume) can start new moves; thin-volume breaks often fail.
4. **Levels flip roles:** broken resistance becomes support, and vice versa. Beware false breakouts — wait for confirmation.

SELF-CHECK

1. Define support and resistance.
2. Why does resistance often form at a price where many people previously bought?
3. What makes a breakout believable versus likely to fail?
4. Explain “role reversal” of a broken level.

II

PART TWO

Candlesticks — the Heart of It

Candlestick charts, invented by Japanese rice traders centuries ago, are the technician's native language. Each candle is a tiny story of one period's battle between buyers and sellers; strung together, they reveal the crowd's mood. This is the most practical part of the book.

ANATOMY | SINGLE-CANDLE | MULTI-CANDLE | CONTEXT

CHAPTER FIVE

The Anatomy of a Candlestick

One candle holds four prices and an entire story of conflict. Learn to read a single candle and you can read the market's mood one heartbeat at a time.

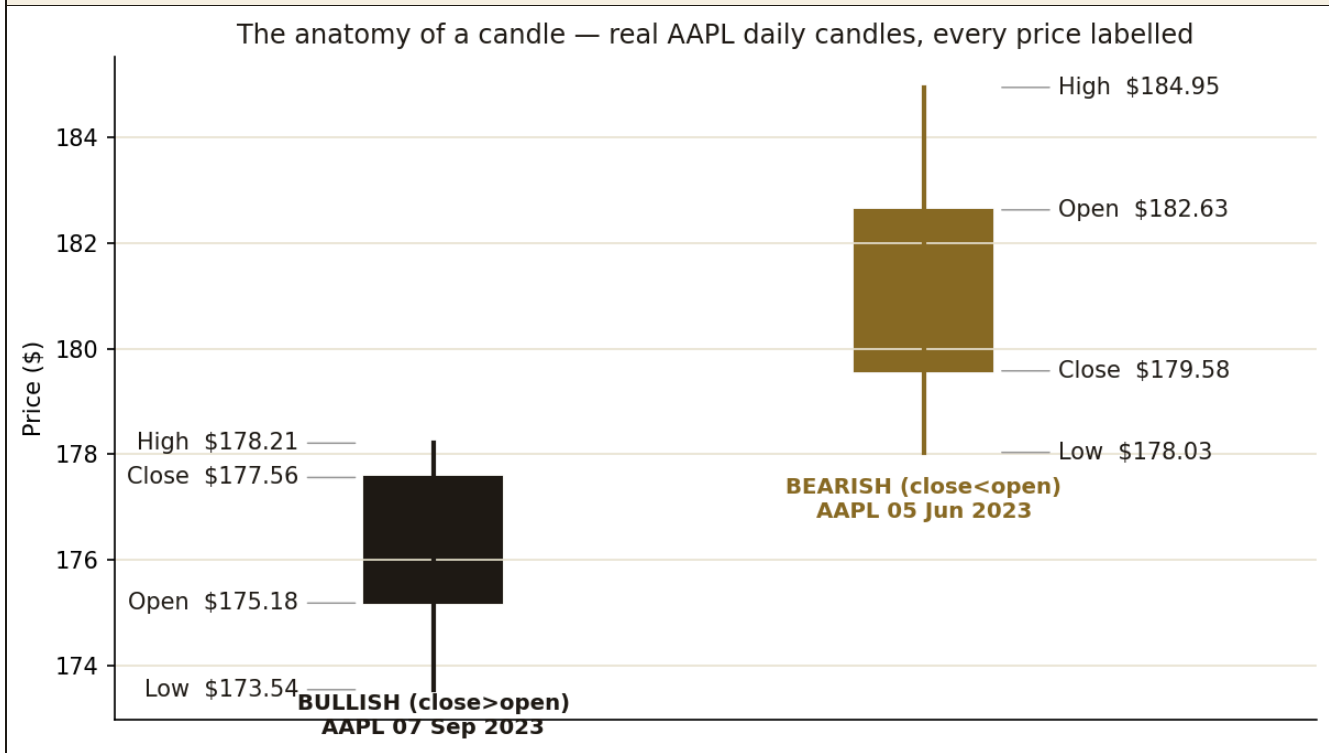
IN THIS CHAPTER YOU WILL LEARN

- The four prices every candle encodes — open, high, low, close.
- The parts of a candle: body, wicks, and colour.
- What the size of body and wick tells you about the battle.

A candlestick is the most information-dense symbol in finance: four numbers and a colour, packed into a shape your eye reads in an instant. Master its anatomy here and every pattern in the chapters ahead becomes self-explanatory.

5.1 Four Prices, One Candle

Each candle summarises one period — a day, an hour, a minute, whatever your time frame. It records four prices: the **open** (first trade of the period), the **high** (highest price reached), the **low** (lowest), and the **close** (last trade). From just these four numbers the candle is drawn — and from its shape you read who won the period: buyers or sellers.

FIGURE 5.1 · THE ANATOMY OF A CANDLE — REAL APPLE CANDLES

Read it like a thermometer of mood. Two real AAPL daily candles, every price labelled. The body spans the open and close; the wicks reach the high and low. Here a **dark** candle is bullish (close above open) and a **gold** candle bearish (close below open).

5.2 Body, Wick & Colour

The **body** is the thick rectangle between the open and the close. If price closed *higher* than it opened, it's a bullish (up) candle; if it closed *lower*, it's bearish (down). The **wicks** (also called shadows or tails) are the thin lines above and below the body, marking the high and low the price touched before settling. Colour tells you direction at a glance; the shape tells you the story.

5.3 What the Sizes Mean

The proportions carry the message. A **long body** means the close was far from the open — one side dominated decisively (a long up-body = strong buying). A **short body** means open and close were close

together — indecision; neither side won. A **long wick** shows price travelled far in that direction but was *rejected* and pushed back before the close — a long lower wick, for instance, means sellers drove price down but buyers fought it back up, a quietly bullish sign. Reading a candle, then, is reading a tug-of-war: who reached furthest, and who was holding the rope when the period ended.

In Plain Terms · the candle as a sentence

Think of each candle as a one-sentence news report. “Buyers pushed hard and closed near the high” = long dark body, tiny wicks. “Sellers tried to crash it but buyers rescued it” = small body sitting atop a long lower wick. You're not memorising shapes; you're translating a battle into a sentence. Once you can narrate candles like this, the famous “patterns” of the next chapters are just two- and three-sentence stories.

KEY TAKEAWAYS

1. **Each candle encodes four prices:** open, high, low, close — one period's whole story.
2. **Body = open-to-close; wicks = the high and low.** Colour shows who won (here: dark up, gold down).
3. **Long body = decisive; short body = indecision; long wick = a rejected move.**
4. **Reading a candle is reading a tug-of-war** — narrate it as a sentence and patterns become obvious.

SELF-CHECK

1. Name the four prices a candle records.
2. What's the difference between the body and the wicks?
3. What does a long lower wick suggest about the period's battle?
4. You see a tiny body with long wicks on both sides — what mood does that signal?

CHAPTER SIX

Single-Candle Patterns

Some candles are so distinctive they carry meaning on their own. The doji, the hammer, the shooting star — each is a single candle that hints the mood may be shifting. Here are the ones worth knowing.

IN THIS CHAPTER YOU WILL LEARN

- The doji — and what indecision looks like.
- The hammer and hanging man — long lower wicks.
- The shooting star and inverted hammer — long upper wicks.
- The marubozu — total dominance.

A single candle rarely makes a decision for you, but a few shapes are loud enough to demand attention — especially when they appear after a long trend. Learn these four families and you'll recognise most of what single candles can say.

6.1 The Doji — Indecision

A **doji** is a candle with almost *no body* — the open and close are nearly equal — leaving a cross or plus-sign shape. It means the period was a draw: buyers and sellers fought to a standstill. By itself a doji is neutral, but after a strong trend it's a warning that the trend's momentum may be fading — the crowd has lost conviction. A doji after a long rally, especially, says “the buyers are tiring.”

6.2 Hammer & Hanging Man

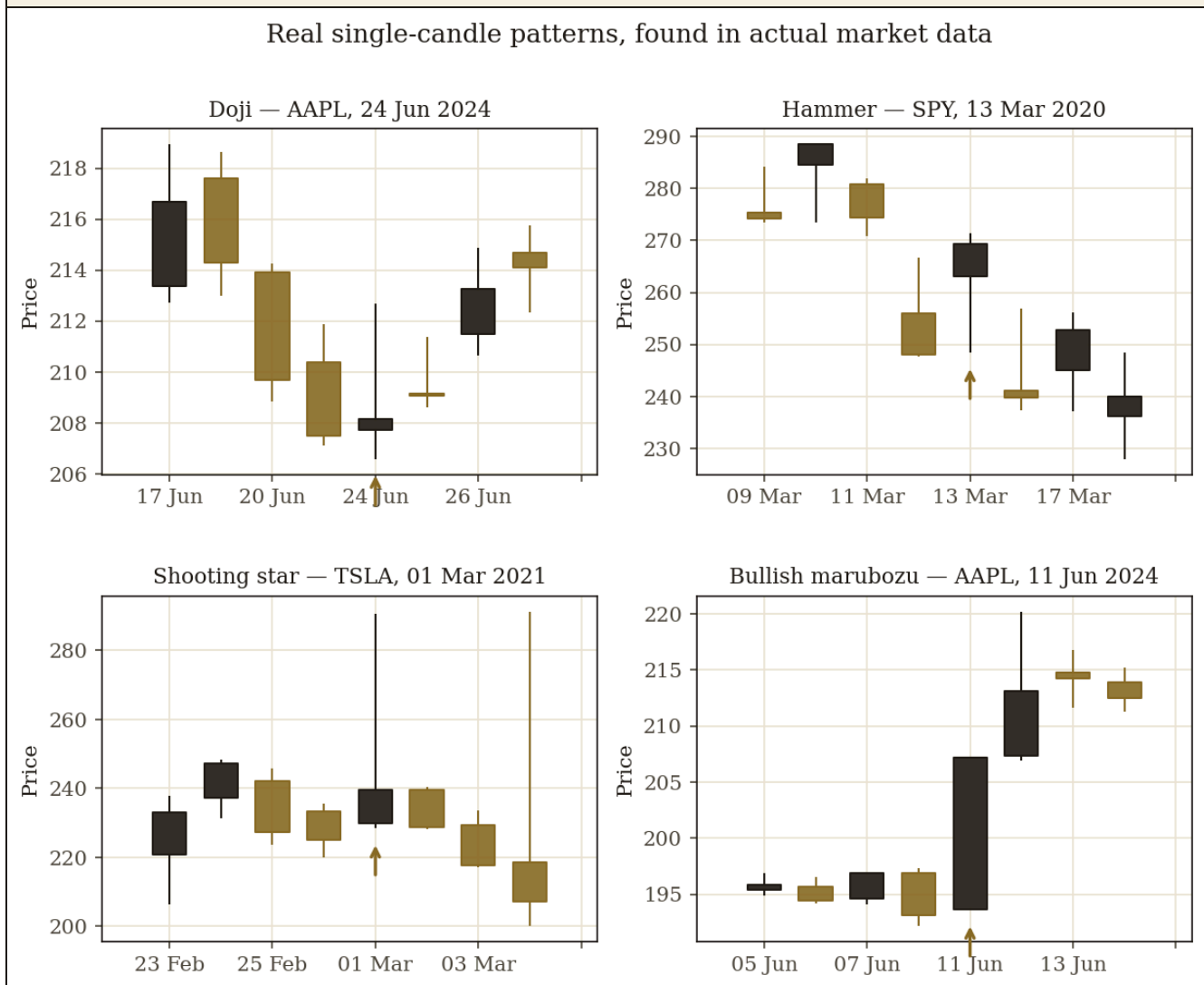
Both have a **small body at the top and a long lower wick** — price was driven sharply down, then rescued back up before the close. The same shape carries opposite meaning depending on *where* it appears (a preview of Chapter 8's great lesson). After a *downtrend*, it's a **hammer** — bullish, suggesting sellers tried to push lower but buyers slammed the door; the decline may be ending. After an *uptrend*, the identical shape is a **hanging man** — bearish, a hint that selling pressure is creeping in beneath a still-rising price.

6.3 Shooting Star & Inverted Hammer

The mirror image: a **small body at the bottom and a long upper wick** — price surged up but was rejected and fell back. After an *uptrend*, this is a **shooting star** — bearish, showing buyers pushed to new highs but sellers overwhelmed them; the rally may be exhausted. After a *downtrend*, the same shape is an **inverted hammer** — a tentative bullish hint.

6.4 The Marubozu — Total Control

A **marubozu** is a long body with *no wicks* at all — price opened at one extreme and closed at the other, one side in complete control for the whole period. A bullish (dark) marubozu shows relentless buying; a bearish (gold) one, relentless selling. It signals strong conviction and momentum in that direction.

FIGURE 6.1 · REAL SINGLE-CANDLE PATTERNS, FOUND IN MARKET DATA

Real, not drawn. Four genuine examples scanned from actual market data: a doji (indecision), the SPY COVID-crash hammer, TSLA's dramatic shooting star, and an AAPL bullish marubozu. The arrow marks each pattern candle; the date lets you verify it.

WATCH OUT

One candle is a hint, not a command

Beginners memorise these shapes and then trade every doji or hammer they see — and lose. A single candle is a *whisper*, easily wrong on its own. The professional waits for **confirmation**: the next candle moving in the expected direction, ideally on good volume, and the pattern appearing at a sensible spot (a support level, the end of a long trend). A hammer in the middle of nowhere means little. Never act on one candle alone.

KEY TAKEAWAYS

1. **Doji** = near-zero body = indecision; after a trend, a warning momentum is fading.
2. **Hammer / hanging man** = small body, long lower wick — bullish after a fall, bearish after a rise.
3. **Shooting star / inverted hammer** = small body, long upper wick — bearish after a rise, bullish after a fall.
4. **Marubozu** = full body, no wicks = total control. And always wait for confirmation.

SELF-CHECK

1. What does a doji look like, and what does it mean after a long rally?
2. A candle has a small body on top and a long lower wick, appearing after a downtrend. Name it and its meaning.
3. How can the hammer and hanging man be the same shape yet mean opposite things?
4. Why should you never act on a single candle alone?

CHAPTER SEVEN

Multi-Candle Patterns

When two or three candles combine, the story gets stronger. The engulfing pattern, the star, the three-soldiers — these multi-candle formations are the bread-and-butter signals technicians actually watch for.

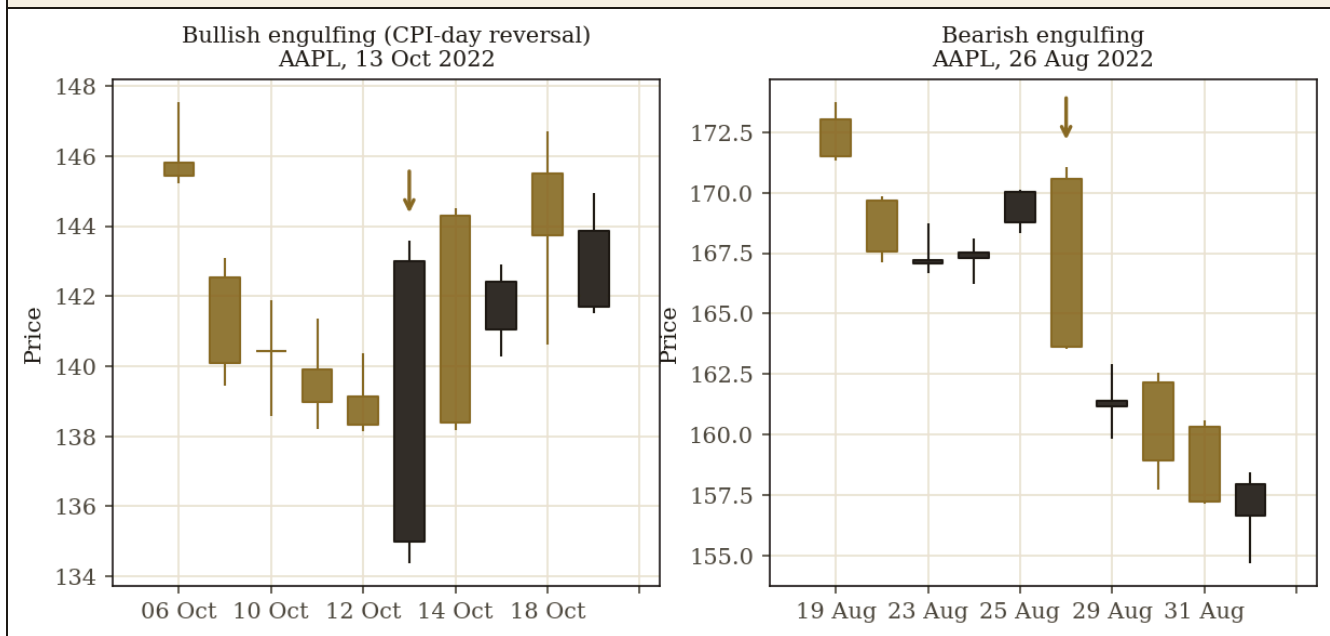
IN THIS CHAPTER YOU WILL LEARN

- The engulfing patterns — bullish and bearish.
- The morning and evening star.
- Three white soldiers and three black crows.
- Harami and piercing patterns.

A single candle whispers; a well-formed multi-candle pattern speaks. Because these patterns require two or three candles to *agree* on a story, they're more reliable than any lone candle — though, as ever, confirmation and context still rule.

7.1 Engulfing Patterns

The **bullish engulfing** is two candles: a small down candle, then a large up candle whose body completely “engulfs” it. After a downtrend it's a strong reversal signal — buyers didn't just win the second period, they overwhelmed the previous selling. The **bearish engulfing** is the mirror: after an uptrend, a small up candle is swallowed by a large down candle, signalling sellers have seized control. Engulfing patterns are among the most watched reversal signals precisely because the second candle's dominance is so visible.

FIGURE 7.1 · REAL ENGULFING PATTERNS — APPLE

Engulfing = a takeover. Left: a real bullish engulfing — the 13 Oct 2022 CPI-day reversal, where a large up-candle swallowed the prior down-candle. Right: a real bearish engulfing. The arrow marks the engulfing candle in each.

7.2 Morning & Evening Star

These three-candle patterns are classic reversals. The **morning star** (bullish) appears at the bottom of a downtrend: a big down candle, then a small indecisive candle (a “star”, often a doji) that gaps lower, then a big up candle — like dawn after a dark night, the mood turns from selling to buying. The **evening star** is the mirror at a top: big up candle, small star, then a big down candle — dusk falling on a rally. The middle “star” is the hinge: it marks the moment of indecision where control changed hands.

7.3 Three Soldiers & Three Crows

Three white soldiers are three strong up candles in a row, each closing higher — a powerful sign of sustained buying, often marking the start of an uptrend after a bottom. **Three black crows** are the bearish mirror: three strong down candles in a row, warning of sustained selling. Both gain their force from *persistence* — three periods of one side winning is hard to dismiss as noise.

7.4 Harami & Piercing

The **harami** (“pregnant” in Japanese) is engulfing in reverse: a large candle followed by a *small* candle held within the first's body — a sign the prevailing trend is losing steam, momentum suddenly contracting. The **piercing pattern** (bullish) and its mirror the **dark cloud cover** (bearish) are two-candle reversals where the second candle pushes more than halfway into the first's body — a partial engulfing that hints the tide is turning. You needn't memorise every name; what matters is reading the *story* each tells about shifting control.

In Plain Terms · don't drown in names

There are dozens of named candle patterns, and beginners often try to memorise them all like flashcards. Don't. Every one of them is just a small story about *who is winning and whether that's changing*. If you understand the anatomy (Chapter 5) and can narrate a candle as a sentence, you can *derive* the meaning of any pattern on sight — engulfing is “a takeover,” a star is “a turning point,” three soldiers is “relentless buying.” Understand the logic; the vocabulary follows.

KEY TAKEAWAYS

1. **Engulfing:** a large candle swallows the prior one — a visible takeover; reversal signal after a trend.
2. **Morning/evening star:** three candles with an indecisive “star” hinge marking the turn.
3. **Three soldiers / three crows:** three candles in a row showing persistent buying or selling.
4. **Read the story, don't memorise names** — every pattern is about who's winning and whether it's changing.

SELF-CHECK

1. Describe a bullish engulfing pattern and where it's meaningful.

2. What are the three candles of a morning star, and what does the middle one represent?
3. Why are three white soldiers considered a strong signal?
4. How is a harami the “opposite” of an engulfing pattern in meaning?

CHAPTER EIGHT

Reading Candles in Context

The single most important lesson in candlestick analysis is also the most ignored: a candle means nothing by itself. The same shape is bullish in one place and bearish in another. Context is everything.

IN THIS CHAPTER YOU WILL LEARN

- Why location changes a candle's meaning entirely.
- How to combine candles with trend, support/resistance and volume.
- What “confirmation” really means in practice.

You now know the candle shapes. This chapter is the antidote to the most common way they're misused — treating a shape as a magic signal regardless of where it appears. Get this chapter right and you'll avoid most beginner candle mistakes.

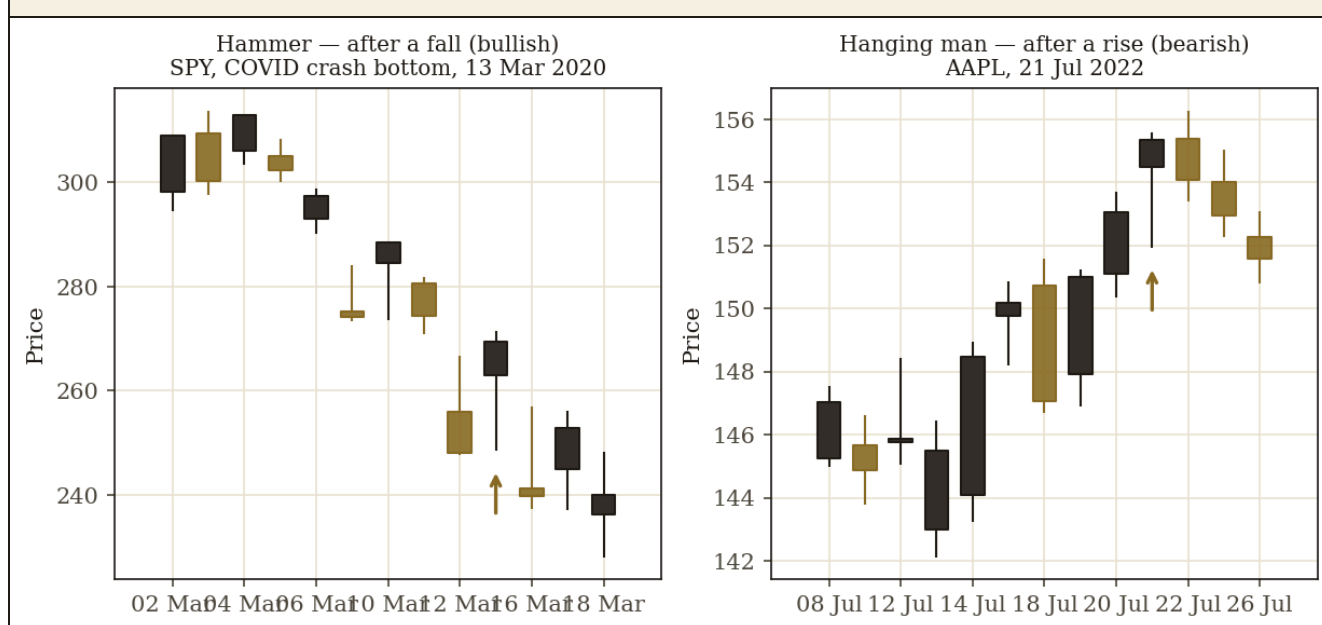
8.1 Location Is Meaning

Recall the hammer and hanging man (Chapter 6): identical shapes, opposite meanings, decided entirely by the trend that preceded them. This generalises to *every* pattern. A doji in the middle of a quiet range is noise; the same doji at the top of a euphoric rally is a serious warning. A bullish engulfing is powerful at a major support level after a long decline; the same pattern mid-trend is barely worth noting. **The candle gives you a shape; the location gives you the meaning.** Always ask first: *where are we?* — what's the trend, and are we at a significant level?

8.2 Combine With Trend, Levels & Volume

The strongest signals occur when several pieces of evidence *align*. A bullish reversal candle is most trustworthy when it appears (a) after a clear downtrend, (b) right at a known support level (Chapter 4), and (c) on heavy volume (Chapter 2). That confluence — pattern + trend context + level + volume — is what professionals actually wait for, and it's far rarer (and far better) than the constant stream of isolated shapes a beginner trades. One weak signal is noise; three signals agreeing is a setup.

FIGURE 8.1 · SAME SHAPE, OPPOSITE MEANING — REAL EXAMPLES



Location is meaning. The identical long-lower-wick candle is a bullish **hammer** after a fall (left — the SPY COVID-crash bottom) and a bearish **hanging man** after a rise (right). Same shape; only the context differs. That is the whole lesson of this chapter.

8.3 What Confirmation Really Means

Throughout these chapters we've said "wait for confirmation." Concretely, it means not acting on a pattern until *subsequent* price action agrees with it. A bullish engulfing isn't confirmed until the *next* candle also closes higher; a breakout isn't confirmed until price *closes* beyond the level (not just pokes through intraday) and holds. Confirmation costs you a little — you enter slightly later, at a slightly worse price — but it filters out a huge share of false signals. That trade-off, accepting a worse entry to avoid

being faked out, is one of the quiet disciplines that separates those who survive from those who churn their accounts to zero.

WATCH OUT

The pattern-spotting trap

Your brain is a pattern-detection machine — so good at it that it sees patterns in pure randomness (clouds, coffee stains, and yes, price charts). A great deal of what beginners “see” in charts is their own mind imposing a story on noise. The defences in this chapter — demanding context, confluence and confirmation — exist precisely to discipline that instinct. If a setup requires you to squint and tell yourself a story to see it, it isn't there. The best setups are obvious; if you have to talk yourself into it, walk away.

KEY TAKEAWAYS

1. **A candle means nothing alone** — the same shape is bullish or bearish depending on location.
2. **Always ask “where are we?”** — trend and level decide a pattern's meaning.
3. **The best signals show confluence:** pattern + trend context + key level + volume agreeing.
4. **Confirmation** = waiting for later price action to agree; it costs a little and saves a lot.

SELF-CHECK

1. Why does the location of a candle change its meaning? Give an example.
2. What four things, when they align, make a reversal signal strongest?
3. What does it mean, concretely, to wait for confirmation of a bullish engulfing?
4. Why is the human tendency to see patterns a danger in chart-reading?

III

PART THREE

Chart Patterns

Zoom out from individual candles and larger shapes emerge across weeks and months — heads and shoulders, triangles, flags. These chart patterns are the macro-version of candle reading: the crowd's psychology drawn on a bigger canvas.

REVERSALS | CONTINUATIONS | GAPS & VOLUME

CHAPTER NINE

Reversal Patterns

Some chart shapes tend to appear when a trend is exhausting itself and about to turn. The head-and-shoulders, the double top and bottom — these reversal patterns are among the most famous (and most over-claimed) ideas in technical analysis.

IN THIS CHAPTER YOU WILL LEARN

- The head-and-shoulders top (and its inverse).
- Double tops and double bottoms.
- The role of the neckline and volume.
- An honest word on how reliable these really are.

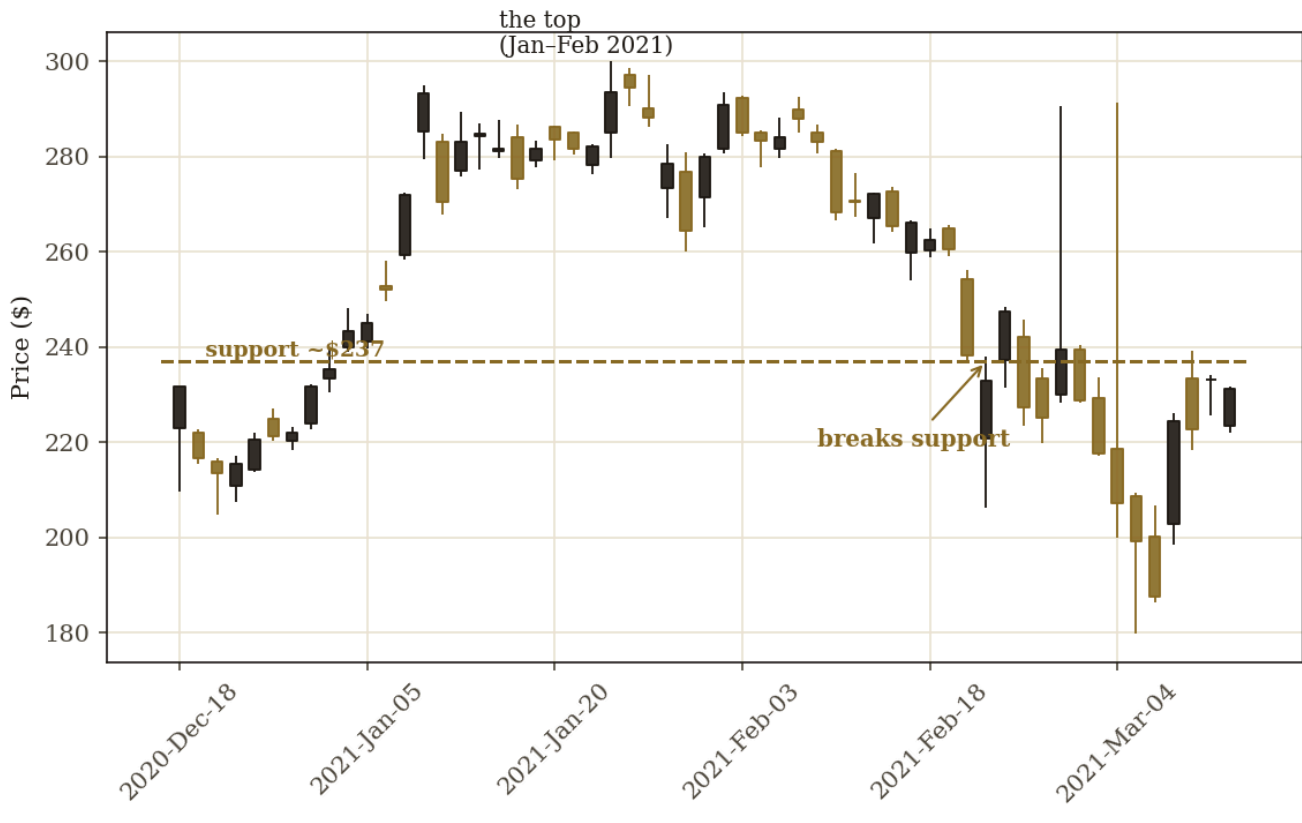
A reversal pattern is a shape that forms as a trend runs out of energy and prepares to change direction. They're seductive because they seem to call the top or bottom — but they only “work” a fraction of the time, and only with confirmation. Learn them, and learn their limits.

9.1 Head and Shoulders

The **head-and-shoulders top** is the most famous reversal pattern. After an uptrend, price makes a peak (left shoulder), a higher peak (the head), then a lower peak (right shoulder) — three humps, the middle tallest. A line connecting the lows between them is the **neckline**. When price breaks *below* the neckline, the pattern is considered complete and a downtrend may begin. The **inverse head-and-shoulders** is the mirror at a market bottom, signalling a possible move up. The logic: each failed push to a new high shows buyers weakening, until they finally give way.

FIGURE 9.1 · A REAL TOPPING PATTERN — TESLA, EARLY 2021

A real topping pattern — TSLA, early 2021: a rounded top, then a break of support



Real tops are messy. Real reversals rarely look like the textbook. TSLA's early-2021 rounded top rolled over and then broke its ~\$237 support — the break being the actual signal. Compare this to the neat diagrams elsewhere: this is exactly Chapter 9's warning, drawn from real data.

9.2 Double Tops & Bottoms

A **double top** looks like the letter “M”: price rises to a peak, pulls back, rises to roughly the *same* peak again, then fails and falls — twice rejected at the same level (a resistance, Chapter 4), the buyers give up. A **double bottom** is the “W” mirror at a market low: price tests the same support twice, holds, and turns up. Like head-and-shoulders, the pattern isn't “confirmed” until price breaks the level between the two humps (the middle of the M or W).

9.3 The Role of the Neckline & Volume

Across all reversal patterns, two features matter most. The **break of the key level** (neckline, or the midpoint of a double top) is the actual signal — the pattern shape alone is just a setup. And **volume** should ideally confirm: a genuine head-and-shoulders often shows declining volume on the head and right shoulder (waning enthusiasm) and a surge of volume on the neckline break. Without that volume confirmation, treat the pattern with extra suspicion.

WATCH OUT · THE HONEST TRUTH ABOUT THESE PATTERNS

They fail far more often than the books admit

Reversal patterns are presented in many courses as near-magical predictors. The reality, from serious studies of pattern performance, is humbler: they work only somewhat better than chance, frequently fail (the “textbook” head-and-shoulders that simply keeps going up), and are easy to *see in hindsight* but hard to trade in real time. Treat them as one piece of probabilistic evidence — never as a guarantee — and always pair them with confirmation, a stop-loss (Chapter 17), and the humility of Chapter 1. Anyone selling them as certainties is selling you something.

KEY TAKEAWAYS

1. **Head-and-shoulders:** three peaks (middle highest); a break below the neckline signals a possible top.
2. **Double top (M) / double bottom (W):** a level tested twice and rejected, signalling reversal on the break.
3. **The break of the key level is the signal**, not the shape; volume should confirm it.
4. **Be honest:** these patterns fail often and are clearer in hindsight — evidence, not prophecy.

SELF-CHECK

1. Describe the three peaks of a head-and-shoulders top and what completes the pattern.
2. What does a double top look like, and what crowd behaviour does it reflect?

3. What is the neckline, and why is its break the real signal?
4. Give one honest limitation of reversal patterns.

CHAPTER TEN

Continuation Patterns

Not every pattern means a turn. Some are pauses — moments where a trend catches its breath before continuing. Triangles, flags and pennants are the market resting, not reversing.

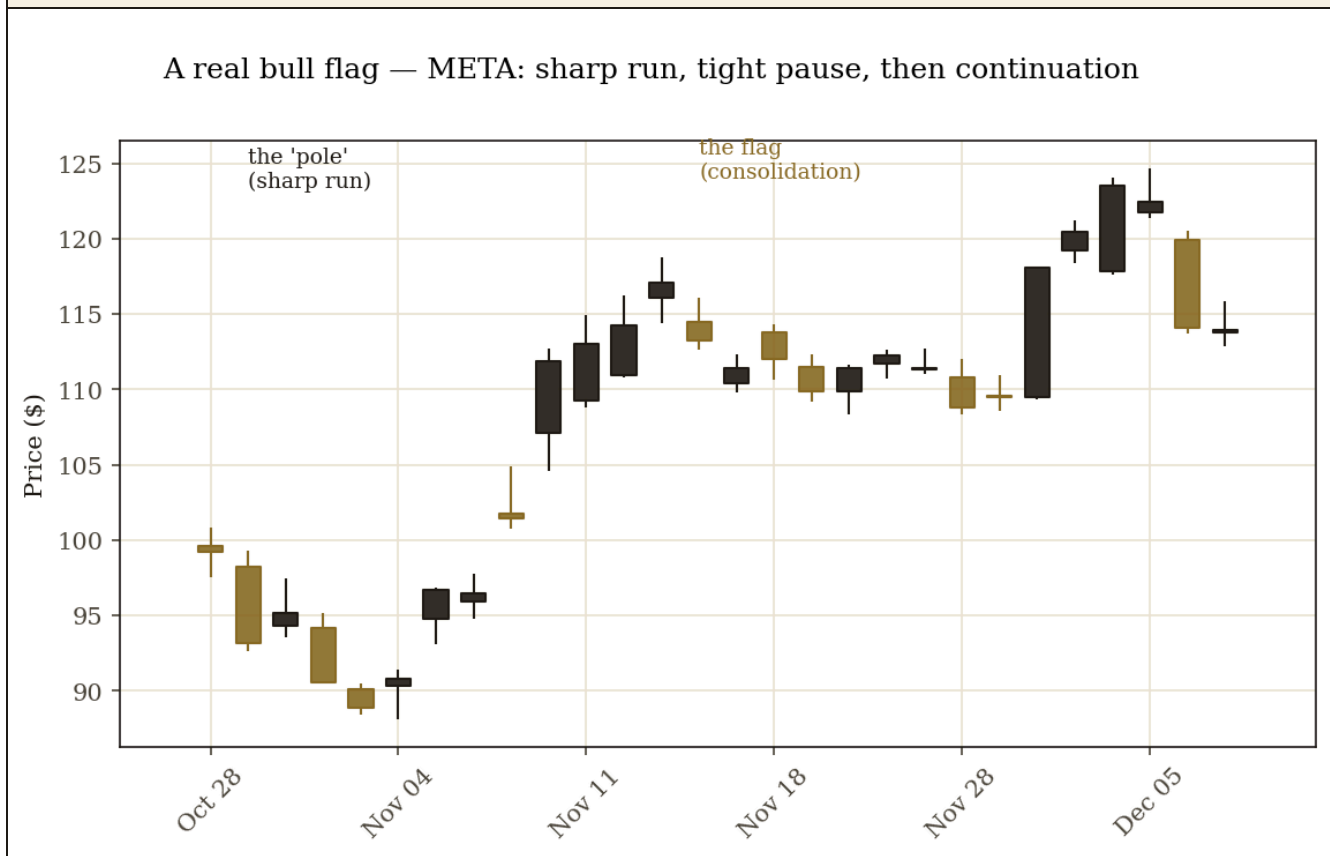
IN THIS CHAPTER YOU WILL LEARN

- Triangles — ascending, descending, symmetrical.
- Flags and pennants — the brief pause.
- How to tell a pause from a reversal.

A continuation pattern is a consolidation — a sideways breather within a larger trend. After it, the trend usually resumes in its original direction. Recognising these helps you stay in a good trend instead of bailing out during a normal pause.

10.1 Triangles

A **triangle** forms as price swings narrow into a point, like a coiling spring. An **ascending triangle** has a flat top (resistance) and rising lows — buyers growing more eager, usually breaking out upward. A **descending triangle** has a flat bottom and falling highs — typically breaking down. A **symmetrical triangle** narrows from both sides and can break either way (so wait for the break to show direction). The squeeze reflects tension building between buyers and sellers until one side wins.

FIGURE 10.1 · A REAL BULL FLAG — META, LATE 2022

The trend catching its breath. Real META data: a sharp run up (the 'pole'), a tight sideways pause (the 'flag'), then the uptrend resumes — a textbook continuation pattern, in real prices.

10.2 Flags & Pennants

A **flag** appears after a sharp move (“the flagpole”): price drifts sideways or slightly against the trend in a small rectangle, then resumes in the original direction. A **pennant** is similar but the pause forms a tiny triangle. Both represent a brief consolidation — traders catching their breath, some taking profits — before the dominant trend reasserts itself. They're typically short-lived (a few periods) and, when they resolve, often do so quickly.

10.3 Pause vs. Reversal

The practical challenge is telling a continuation (pause) from a reversal (turn) — they can look similar while forming. Three clues help: the *prior trend* (continuations happen *within* a strong trend; reversals at its *end*), the *direction of the eventual break* (a continuation breaks *with* the trend), and *volume* (continuations often see volume dry up during the pause, then surge on the resumption). When unsure — and you often will be — the disciplined answer is to wait for the break and let price tell you, rather than guessing.

In Plain Terms · why pauses happen

A strong move makes some holders nervous — they take profits, creating a little selling that stalls the advance. Meanwhile new buyers, who missed the first move, wait for a dip. For a while these two groups balance out and price goes sideways: the flag or triangle. Once the profit-takers are done and the trend's believers remain, the move resumes. The pattern is just the visible shape of that brief tug-of-war within an ongoing trend.

KEY TAKEAWAYS

1. **Continuation patterns are pauses**, after which the prior trend usually resumes.
2. **Triangles** coil into a point: ascending (bullish), descending (bearish), symmetrical (wait for the break).
3. **Flags and pennants** are brief consolidations after a sharp move (the “flagpole”).
4. **Distinguish pause from reversal** via prior trend, break direction, and volume — or just wait for the break.

SELF-CHECK

1. What does a continuation pattern signal, versus a reversal pattern?
2. How do ascending and descending triangles typically resolve?
3. Describe a bull flag and what each part represents.

4. Name two clues that help you tell a pause from a reversal.

CHAPTER ELEVEN

Gaps & Volume

Two raw forces underlie every pattern: the gaps where price leaps with no trades in between, and the volume that reveals how much conviction is behind a move. Master these and you read the energy of the market, not just its shape.

IN THIS CHAPTER YOU WILL LEARN

- What a gap is and why it forms.
- The main types of gap and what they suggest.
- How volume confirms (or contradicts) price.

We met volume in Chapter 2 as the technician's lie-detector; here we deepen it and add its dramatic cousin, the gap. Both are about *energy* — the force behind a move — and both are among the more genuinely informative things on a chart.

11.1 What a Gap Is

A **gap** is an empty space on the chart where price jumps from one period's close to the next period's open with no trading in between. It happens most often between sessions: news breaks overnight — an earnings surprise, a takeover, a shock — and when the market reopens, price simply leaps to a new level, leaving a visible gap. A gap is a burst of pent-up information hitting the price all at once; its size hints at how significant the news was.

11.2 Types of Gap

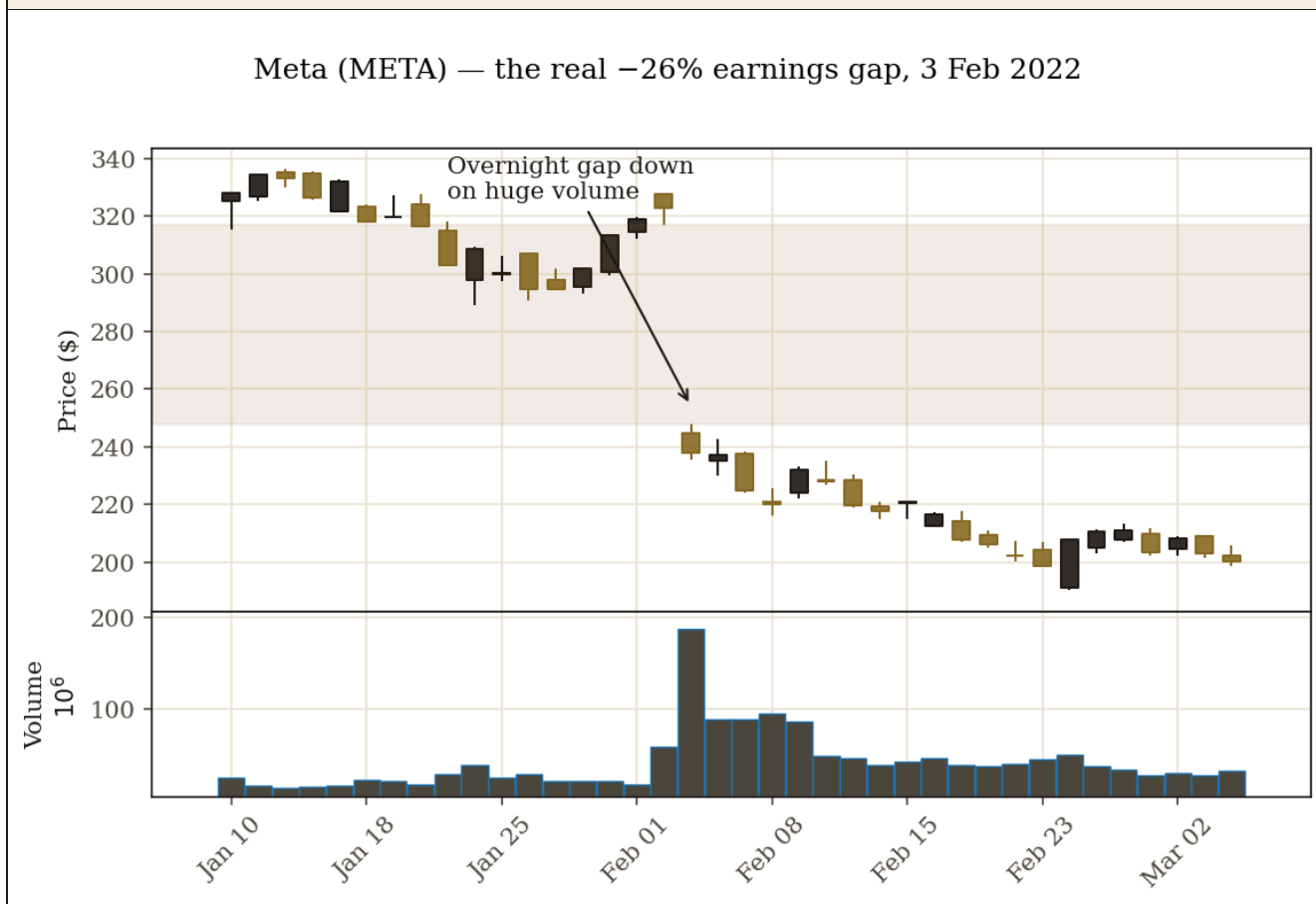
COMPARE & UNDERSTAND · READING THE GAPS

GAP TYPE	WHERE IT APPEARS	WHAT IT SUGGESTS
Breakaway	At the start of a new move, out of a range	A powerful new trend is beginning
Runaway / measuring	In the middle of a strong trend	The trend is healthy and accelerating
Exhaustion	Near the end of an extended trend	A last gasp; the trend may be ending
Common	In quiet, range-bound markets	Little meaning; often “fills” quickly

A famous piece of market lore is that “gaps get filled” — price often returns later to trade through the empty space. It's frequently true for common and exhaustion gaps, but breakaway and runaway gaps in strong trends can stay open for a long time. As always, context (where the gap appears, and on what volume) decides the meaning.

11.3 Volume Confirms Price

The single most useful volume principle bears repeating and expanding: **volume should expand in the direction of the trend**. In a healthy uptrend, volume rises on up-days and fades on pullbacks. A breakout (Chapter 4), a reversal pattern's neckline break (Chapter 9), or a gap is far more trustworthy *on heavy volume*. The most powerful warning sign is **divergence** — price making new highs while volume shrinks, meaning fewer and fewer participants support the advance. That quiet fade in volume often precedes a reversal: the move is running on fumes.

FIGURE 11.1 · A REAL GAP ON HUGE VOLUME — META, FEB 2022

Energy made visible. Meta's -26% earnings gap of 3 Feb 2022 — one of the largest one-day wipeouts in market history. Price leaps down overnight through the shaded zone (no trades occur there), on enormous volume (lower panel). The volume spike is what gives a gap its conviction.

WATCH OUT

Don't chase the gap

A big morning gap is exciting, and beginners often pile in at the open — right as early buyers take profits, causing the price to drop back (sometimes filling the gap) and handing the latecomer an instant loss. Gaps are highly volatile and emotionally charged; jumping in at the open is one of the faster ways to lose money. Let the dust settle, watch whether the gap holds, and remember Chapter 17's stop-losses exist precisely for moments like these.

KEY TAKEAWAYS

1. **A gap** is an empty space where price leaps with no trades between — usually driven by news.
2. **Gap types** (breakaway, runaway, exhaustion, common) suggest different things by their location.
3. **Volume should expand with the trend;** a move on heavy volume is far more trustworthy.
4. **Volume divergence** (price up, volume shrinking) warns a move is running out of energy.

SELF-CHECK

1. What is a gap, and what typically causes one?
2. Distinguish a breakaway gap from an exhaustion gap.
3. What should volume do in a healthy uptrend?
4. What is volume divergence, and why is it a warning?

IV

PART FOUR

Indicators — with the Math

Indicators are formulas applied to price and volume to distil them into a single, easier-to-read line. Here we don't just name them — we show the actual equations and work real numbers through them, so you understand exactly what each one computes and where it lies to you.

MOVING AVERAGES

RSI & MACD

BOLLINGER & ATR

USING THEM WISELY

CHAPTER TWELVE

Moving Averages

The moving average is the most-used indicator in all of technical analysis — a smoothed line that cuts through the noise to show the underlying trend. Here is exactly how it's calculated, by hand, twice.

IN THIS CHAPTER YOU WILL LEARN

- The simple moving average (SMA) — the formula and a worked calculation.
- The exponential moving average (EMA) — and why it reacts faster.
- How crossovers generate signals (the golden & death cross).
- The fatal weakness all moving averages share.

Price is jagged and noisy; a moving average is the same price, smoothed. By averaging the last N closing prices and re-computing every period, it produces a flowing line that makes the trend obvious. It is the foundation on which countless other indicators are built — so let's build it ourselves.

12.1 The Simple Moving Average (SMA)

The **simple moving average** is exactly what it sounds like: the average of the last N closing prices. Each period you drop the oldest price and add the newest, so the average “moves” along.

$$SMA_N = (P_1 + P_2 + \dots + P_N) \div N$$

Let's compute a 5-day SMA on a real little series of closing prices. Suppose a stock closed at **\$20**, **\$22**, **\$21**, **\$24**, **\$23** over five days, then **\$25** on day 6:

WORKED CALCULATION · 5-DAY SMA

Day 1–5 closes: 20, 22, 21, 24, 23

Sum = 20 + 22 + 21 + 24 + 23 = **110**

SMA (day 5) = 110 ÷ 5 = **\$22.00**

Now day 6 closes at 25 – drop the oldest (20), add the newest (25):

New window: 22, 21, 24, 23, 25

Sum = 22 + 21 + 24 + 23 + 25 = **115**

SMA (day 6) = 115 ÷ 5 = **\$23.00**

The average rose from \$22.00 to \$23.00 – the line “moves” up, smoothly tracking the rising prices without the day-to-day jitter.

Common settings are the 50-day and 200-day SMA for long-term trends, and the 20-day for shorter ones. When price is above its rising moving average, the trend is generally up; below a falling average, generally down.

12.2 The Exponential Moving Average (EMA)

The SMA treats a price from N days ago as equally important as yesterday's — which feels wrong, since recent prices matter more. The **exponential moving average** fixes this by weighting recent prices more heavily. It uses a smoothing multiplier:

$$\text{multiplier } k = 2 \div (N + 1)$$

$$\text{EMA}_{\text{today}} = (\text{Price}_{\text{today}} - \text{EMA}_{\text{yesterday}}) \times k + \text{EMA}_{\text{yesterday}}$$

Let's work a 10-day EMA. First the multiplier, then one update step. Say yesterday's 10-day EMA was **\$22.00** and today's close is **\$25.00**:

WORKED CALCULATION · 10-DAY EMA STEP

$$k = 2 \div (10 + 1) = 2 \div 11 = \mathbf{0.1818}$$

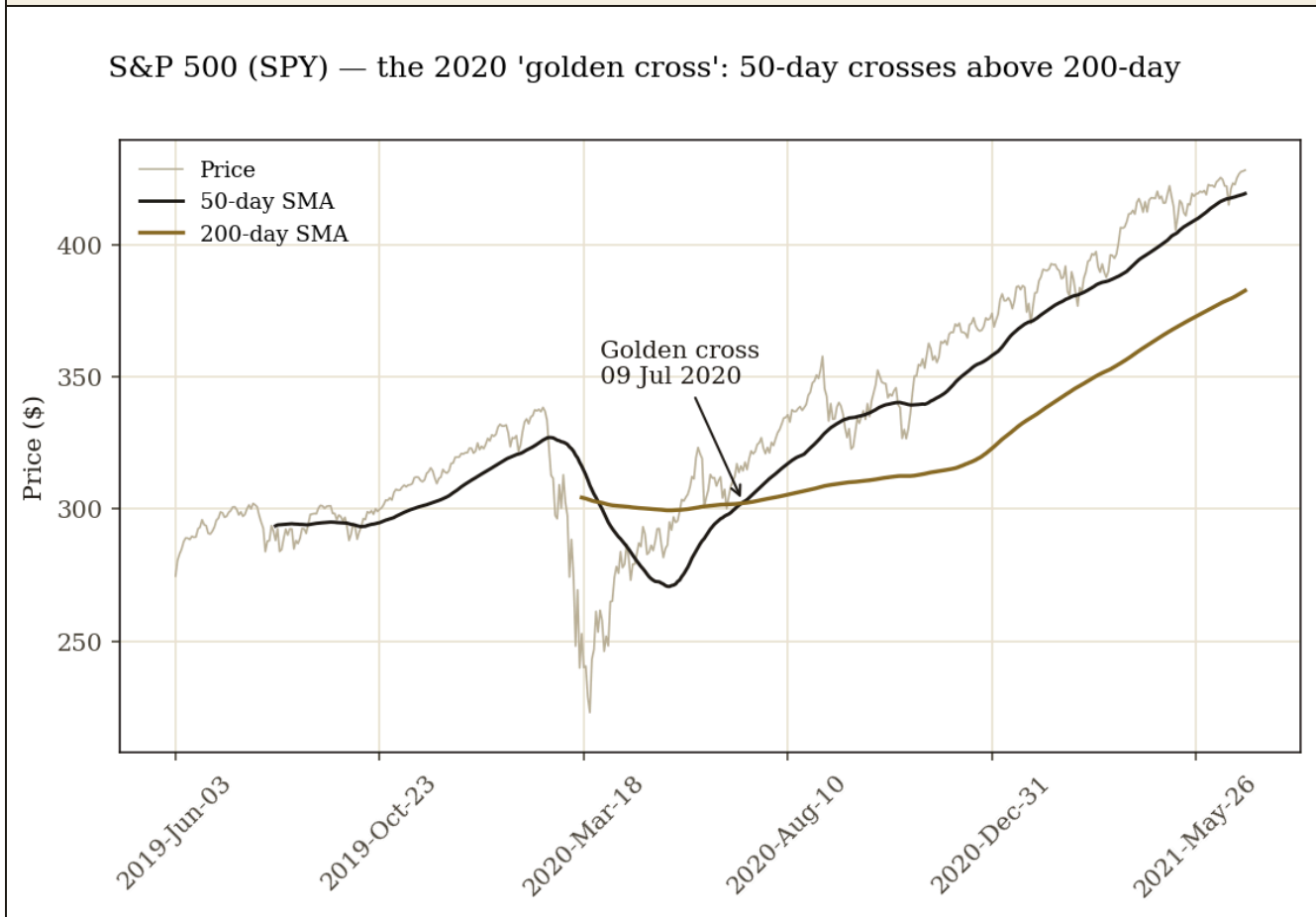
$$\begin{aligned}
 \text{EMA}_{\text{today}} &= (25.00 - 22.00) \times 0.1818 + 22.00 \\
 &= (3.00 \times 0.1818) + 22.00 \\
 &= 0.545 + 22.00 = \text{\$22.55}
 \end{aligned}$$

A 5-day SMA above jumped straight to \$23.00; this EMA moved to \$22.55 – it responds to the new \$25 close but in proportion to its weight (0.1818), giving recent data more pull than old data. Larger N → smaller k → smoother, slower line.

Because it reacts faster, the EMA is preferred by shorter-term traders; the SMA, being smoother, by longer-term ones. Neither is “better” — they’re different trade-offs between responsiveness and noise.

12.3 Crossovers — Golden & Death Cross

A popular signal comes from *two* moving averages of different lengths. When a faster (shorter) average crosses *above* a slower (longer) one, momentum is turning up — a bullish **crossover**. The most famous: the **golden cross**, when the 50-day SMA crosses above the 200-day, taken as a long-term bullish signal; and its opposite, the **death cross** (50-day below 200-day), taken as bearish. These are widely watched — which gives them some self-fulfilling power (Chapter 1) — but they are also *late*, as the next section explains.

FIGURE 12.1 · A REAL GOLDEN CROSS — S&P 500, 2020

Smoothing, and lag. Real SPY data: after the COVID crash, the 50-day moving average (dark) crossed back above the 200-day (gold) on 9 July 2020 — the 'golden cross.' Note how both averages lag the price; the signal arrives well after the actual low.

12.4 The Fatal Weakness — Lag

Every moving average shares one unavoidable flaw: it **lags**. Because it's an average of *past* prices, it always turns *after* the price has already moved. By the time a golden cross appears, a large part of the up-move may be over. This is the central trade-off of all smoothing: a longer average is smoother and more reliable but slower (more lag); a shorter one is faster but noisier (more false signals). There is no setting that escapes this — anyone promising a moving average that predicts turns in advance misunderstands the math. Moving averages describe the trend you're already in; they do not foresee the next one.

WATCH OUT**Whipsaws in a sideways market**

Moving-average crossovers work reasonably in strong trends and *terribly* in sideways, choppy markets — where price keeps crossing back and forth, generating a stream of false buy/sell signals (“whipsaws”) that bleed you through fees and small losses. Before trusting any moving-average signal, ask whether the market is actually trending (Chapter 3). In a range, these tools are worse than useless.

KEY TAKEAWAYS

1. **SMA** = average of the last N closes; recomputed each period. (Worked: 5-day SMA went 22.00 → 23.00.)
2. **EMA** weights recent prices more, via $k = 2/(N+1)$; it reacts faster than the SMA.
3. **Crossovers** (golden/death cross) signal momentum shifts but are widely watched and *late*.
4. **All moving averages lag** and whipsaw in sideways markets — they describe the trend, not the future.

SELF-CHECK

1. Compute the 4-day SMA of closes 10, 12, 14, 16. Then the new value if the next close is 18.
2. What is the EMA multiplier for $N = 9$? Why does the EMA react faster than the SMA?
3. What is a golden cross, and why is it a *lagging* signal?
4. In what kind of market do moving-average crossovers fail badly, and why?

CHAPTER THIRTEEN

RSI & MACD

Two of the most popular momentum indicators, demystified with their actual formulas. The RSI measures how overbought or oversold a stock is; the MACD measures the momentum between two moving averages. We'll compute both by hand.

IN THIS CHAPTER YOU WILL LEARN

- The RSI formula — and a full worked calculation.
- What “overbought” and “oversold” really mean (and don't).
- The MACD — how it's built from EMAs.
- Divergence — the most useful signal these give.

Where moving averages smooth price, momentum indicators measure its *speed* — how fast and forcefully price is moving. The two giants of this family are the RSI and the MACD, and both are far less mysterious once you see the arithmetic.

13.1 The RSI — Formula & Worked Calculation

The **Relative Strength Index** (RSI), created by J. Welles Wilder, measures the magnitude of recent gains against recent losses, on a scale of 0 to 100. The formula:

$$\begin{aligned} \text{RS} &= (\text{average gain}) \div (\text{average loss}) \\ \text{RSI} &= 100 - [100 \div (1 + \text{RS})] \end{aligned}$$

It's usually computed over 14 periods. Let's work a simplified example. Suppose over 14 days the up-days and down-days summed to an **average gain of \$1.00** and an **average loss of \$0.50** per day:

WORKED CALCULATION · 14-DAY RSI

average gain = 1.00 · average loss = 0.50

RS = 1.00 ÷ 0.50 = **2.0**

RSI = 100 - [100 ÷ (1 + 2.0)]

= 100 - [100 ÷ 3]

= 100 - 33.33 = **66.7**

An RSI of 66.7 – gains have outweighed losses 2-to-1, so momentum is firmly up but not yet at the “overbought” 70 line. If average gain equalled average loss, RS = 1 and RSI = 50 (perfect balance).

13.2 Overbought & Oversold — and the Catch

By convention, an RSI **above 70** is called “overbought” (price has risen fast, perhaps due a pullback) and **below 30** “oversold” (fallen fast, perhaps due a bounce). The naïve reading — “sell at 70, buy at 30” — is dangerous, and here's the catch every beginner learns painfully: **in a strong trend, RSI can stay overbought or oversold for a very long time.** A roaring stock can sit above 70 for weeks while it keeps climbing; selling the first time it hit 70 would have cost you the whole run. Overbought is not “sell”; it's “momentum is strong and stretched.” Context (the trend, Chapter 3) decides what to do with that.

13.3 The MACD

The **MACD** (Moving Average Convergence Divergence), built by Gerald Appel, measures the relationship between two EMAs. It has three parts:

$$\text{MACD line} = \text{EMA}_{12} - \text{EMA}_{26}$$

$$\text{Signal line} = \text{EMA}_9 \text{ of the MACD line}$$

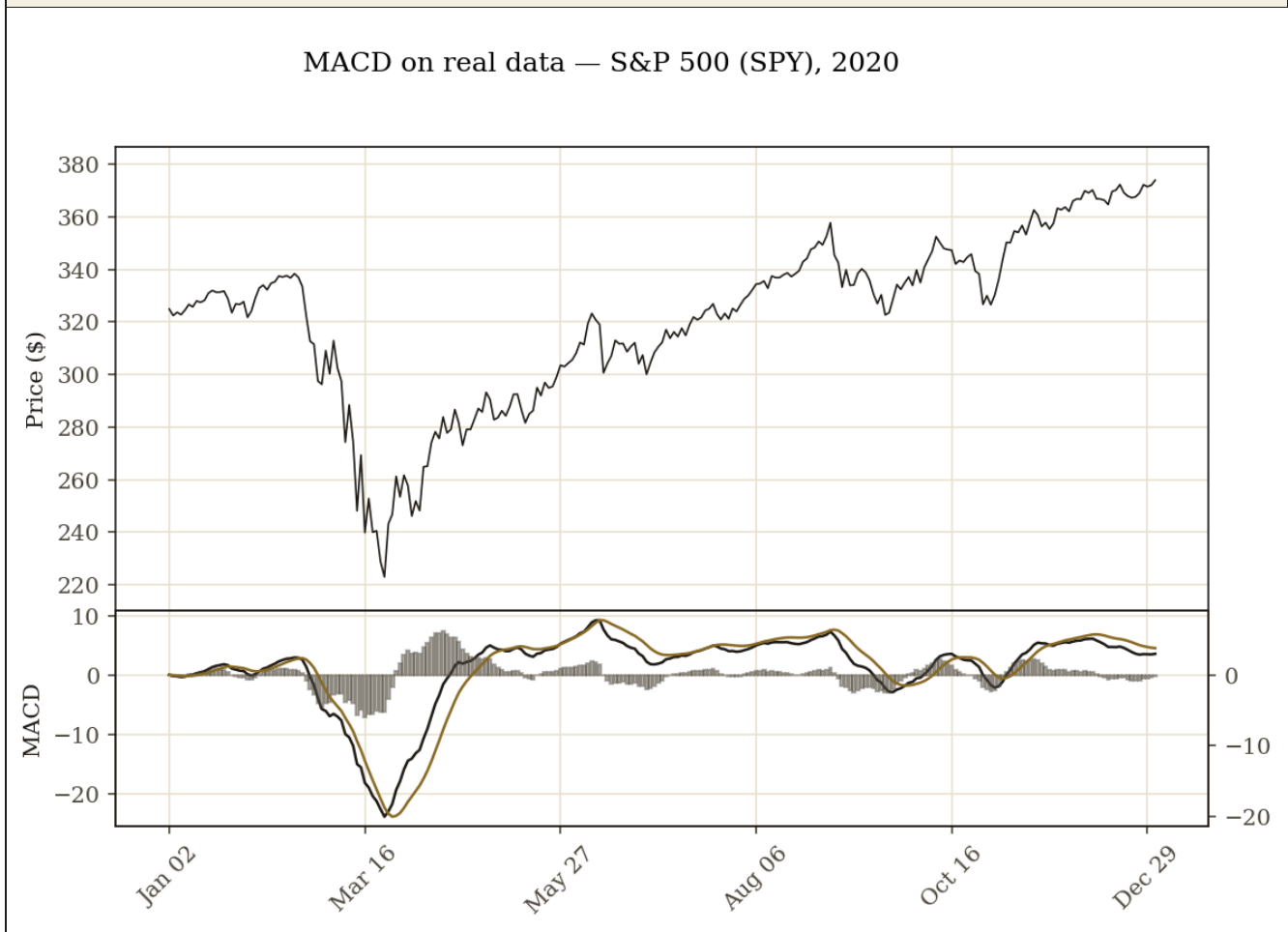
$$\text{Histogram} = \text{MACD line} - \text{Signal line}$$

The MACD line is just the difference between a fast (12-period) and slow (26-period) EMA — so it's positive when the fast average is above the slow (upward momentum) and negative when below. Let's compute the MACD line from two EMAs: say the 12-EMA is **\$51.20** and the 26-EMA is **\$49.80**:

WORKED CALCULATION · MACD LINE

$$\begin{aligned}\text{MACD line} &= \text{EMA}_{12} - \text{EMA}_{26} \\ &= 51.20 - 49.80 = \mathbf{+1.40}\end{aligned}$$

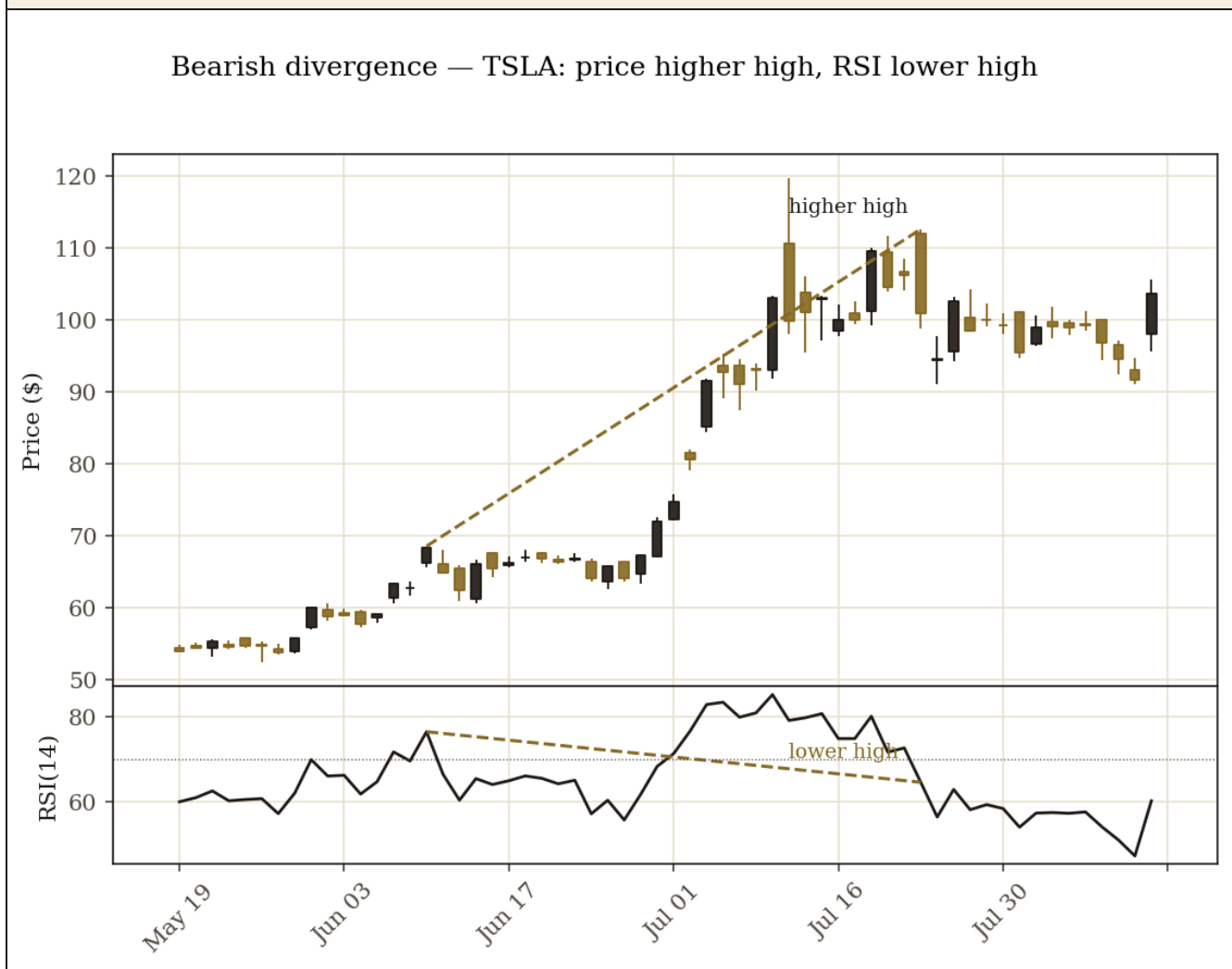
Positive (+1.40) → the fast average is above the slow → upward momentum. When the MACD line crosses above its signal line (the 9-EMA of itself), it's a bullish trigger; crossing below, bearish. The histogram (their difference) shows the momentum growing or shrinking.

FIGURE 13.2 · MACD ON REAL DATA — S&P 500, 2020

The MACD, on real prices. Real SPY data through 2020. The lower panel shows the MACD line (dark) and its signal line (gold); the bars are the histogram of their difference. Crossings of the two lines, and the histogram flipping sign, are the classic momentum triggers — note how they cluster around the real turning points.

13.4 Divergence — the Real Prize

The most genuinely useful signal both indicators give is **divergence** — when the indicator disagrees with price. *Bearish divergence*: price makes a higher high, but RSI (or MACD) makes a *lower* high — the new price peak came with *less* momentum, a warning the uptrend is tiring. *Bullish divergence* is the mirror: price makes a lower low but the indicator makes a higher low, hinting the downtrend is losing force. Divergence is the closest these indicators come to an early warning — because it reveals weakening momentum *beneath* a price that still looks healthy. Even so, like everything in this book, it fails often and needs confirmation.

FIGURE 13.1 · REAL BEARISH DIVERGENCE — TESLA, 2020

The tell beneath the surface. Real TSLA data: price pushes to a higher high while the RSI (lower panel) makes a lower high — momentum quietly weakening beneath a rising price. This divergence is the most prized warning these indicators give, and it is still only a probability.

WATCH OUT

“Overbought” is not a sell signal

The most expensive RSI mistake is reflexively shorting or selling every time RSI crosses 70. In a powerful bull move, RSI can remain pinned above 70 for weeks while the stock doubles — and the trader who “sold the overbought signal” watches the rocket leave without them, or worse, shorts it and gets crushed. Strong momentum *stays* strong longer than feels reasonable. Use these levels as information about momentum, never as automatic commands.

KEY TAKEAWAYS

1. **RSI** = $100 - 100/(1+RS)$, $RS = \text{avg gain} / \text{avg loss}$; 0–100 scale. (Worked: $RS\ 2.0 \rightarrow RSI\ 66.7$.)
2. **Above 70 = overbought, below 30 = oversold** — but in strong trends these persist; not auto buy/sell.
3. **MACD** = $EMA_{12} - EMA_{26}$, with a 9-EMA signal line and a histogram of their difference.
4. **Divergence** (price and indicator disagree) is the most useful early warning — still only probabilistic.

SELF-CHECK

1. If average gain = \$2 and average loss = \$1, compute RS and the RSI.
2. Why is selling the instant RSI hits 70 a costly habit in a strong uptrend?
3. What three components make up the MACD, and what does a positive MACD line mean?
4. Describe bearish divergence and why it's a warning.

CHAPTER FOURTEEN

Bollinger Bands & ATR — Measuring Volatility

The indicators so far measured trend and momentum. These two measure volatility — how much price is bouncing around — which is the key to setting sensible targets and, above all, sensible stop-losses. Again, with the real math.

IN THIS CHAPTER YOU WILL LEARN

- Bollinger Bands — built from a moving average and standard deviation.
- The “squeeze” and what band-width tells you.
- The Average True Range (ATR) — and a worked calculation.
- Why volatility, not prediction, is what these are for.

Volatility is how much price moves, regardless of direction. It matters enormously: the same \$1 move is huge for a calm stock and trivial for a wild one. These two indicators put a number on it.

14.1 Bollinger Bands

Bollinger Bands, devised by John Bollinger, wrap price in a channel built from a moving average and *standard deviation* (a statistical measure of spread). Three lines:

Middle band = 20-period SMA

Upper band = SMA + (2 × standard deviation)

Lower band = SMA − (2 × standard deviation)

Because the bands are set two standard deviations from the average, price stays *inside* them most of the time (statistically, about 95% for normally-distributed data). Let's compute the bands given a 20-day SMA of **\$100** and a standard deviation of **\$4**:

WORKED CALCULATION · BOLLINGER BANDS

Middle band = **\$100.00** (the 20-day SMA)

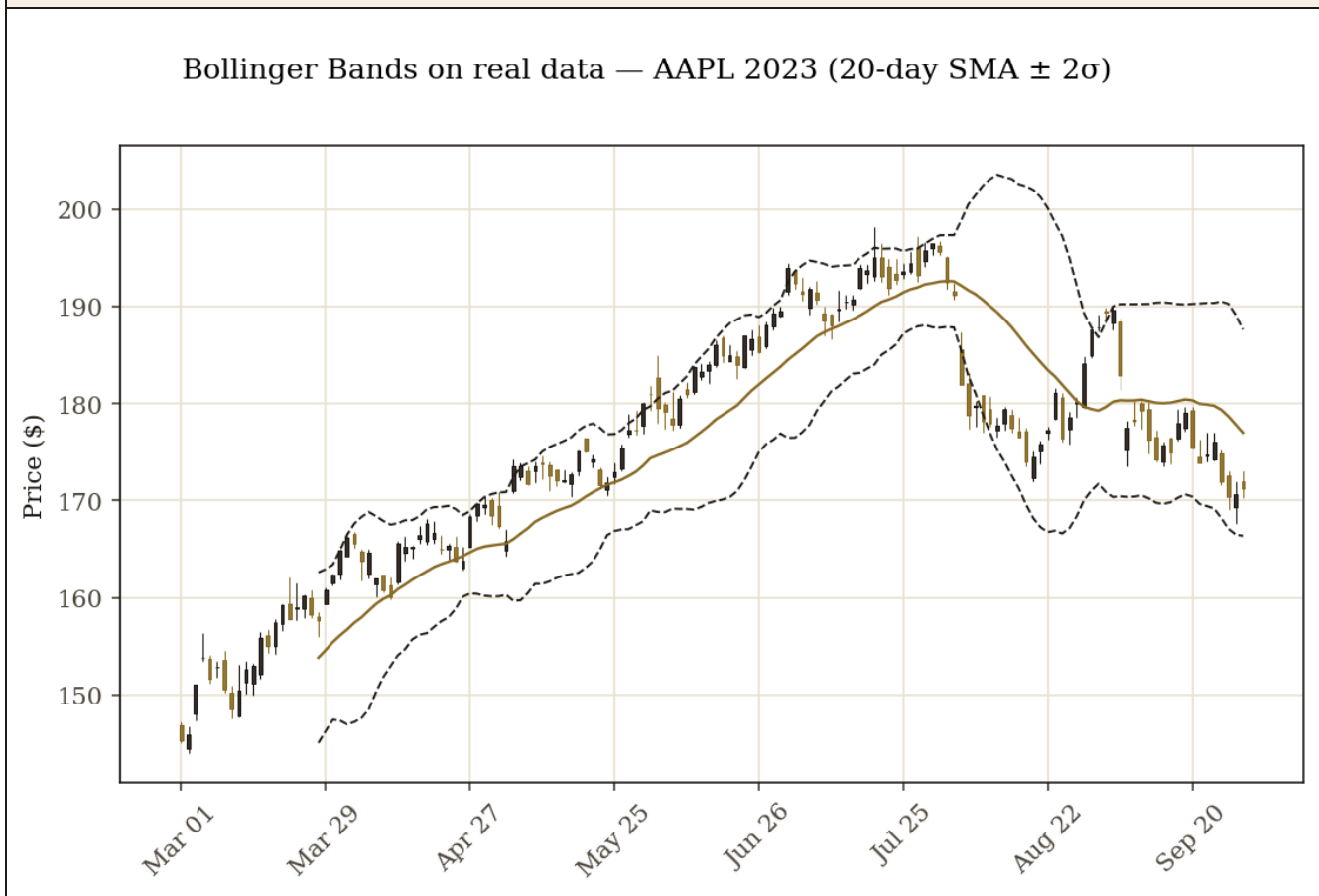
$2 \times \text{std dev} = 2 \times 4 = 8$

Upper band = $100 + 8 = \mathbf{\$108.00}$

Lower band = $100 - 8 = \mathbf{\$92.00}$

Price will spend most of its time between \$92 and \$108. The bands automatically widen when volatility (std dev) rises and pinch in when it falls — so the channel itself is a volatility gauge.

Two common readings: when price tags the upper band it's *relatively* high (and vice versa) — but, exactly like RSI, “riding the band” can persist in a strong trend, so this is not an automatic sell. More useful is the **squeeze**: when the bands pinch very tight, volatility has collapsed — and periods of low volatility tend to be followed by a sharp expansion (a big move). The squeeze doesn't tell you *which way*, only that a move is likely coming.

FIGURE 14.1B · BOLLINGER BANDS ON REAL DATA — APPLE, 2023

The bands breathe with volatility. Real AAPL data: the gold line is the 20-day average, the dashed lines are ± 2 standard deviations. Watch the channel widen when the stock moves sharply and pinch when it calms — the pinch ('squeeze') often precedes a big move, though it never says which way.

14.2 The Average True Range (ATR)

The **Average True Range**, another Wilder creation, measures volatility directly as the average size of a period's price range. First the "true range" (TR) of one period — the greatest of three distances, so that gaps are counted:

$$\text{TR} = \max \text{ of: } (\text{High} - \text{Low}), |\text{High} - \text{prev Close}|, |\text{Low} - \text{prev Close}|$$

$$\text{ATR} = \text{average of TR over } N \text{ periods (usually 14)}$$

Let's compute one period's TR. Today's high is **\$53**, low **\$50**, and yesterday's close was **\$49**:

WORKED CALCULATION · TRUE RANGE, THEN ATR

$$\text{High} - \text{Low} = 53 - 50 = 3$$

$$|\text{High} - \text{prevClose}| = |53 - 49| = 4$$

$$|\text{Low} - \text{prevClose}| = |50 - 49| = 1$$

$$\text{TR} = \max(3, 4, 1) = \$4.00 \quad (\text{the gap up from 49 made the true range bigger than the simple high-low})$$

Do this each day and average over 14 days → the ATR. If the 14-day ATR is, say, \$2.50, this stock typically moves about \$2.50 per day. That single number is gold for stop-losses (next).

14.3 Volatility, Not Prediction

Crucially, neither of these predicts *direction* — they measure *magnitude*. That's exactly why they're so practical for **risk management** (Chapter 17). If a stock's ATR is \$2.50, placing your stop-loss only \$0.50 away is foolish — normal daily noise will stop you out for no reason. Sizing your stop relative to ATR (say, 1.5× or 2× ATR away) means you're giving the trade room to breathe through normal volatility while still capping your loss. Used this way — to *measure risk* rather than to *predict price* — volatility indicators are among the most genuinely useful tools in this entire book.

In Plain Terms · why volatility beats prediction

Most of this book has warned that prediction is unreliable. Volatility indicators are refreshing because they don't pretend to predict — they measure something real and current (how much this thing moves) and let you *size your risk to it*. A trader who can't forecast direction at all can still survive and even thrive by managing volatility well: small positions in wild instruments, stops set beyond the noise, never risking more than a sliver on any one bet. That's the quiet secret of the pros — risk control, not crystal balls.

KEY TAKEAWAYS

1. **Bollinger Bands** = $20\text{-SMA} \pm 2$ standard deviations; price stays inside most of the time. (Worked: \$92–\$108.)
2. A “**squeeze**” (tight bands) signals low volatility, often before a big move — direction unknown.
3. **ATR** averages the “true range” (which counts gaps) to measure how much price moves per period.
4. **These measure magnitude, not direction** — their best use is sizing stop-losses to a stock's volatility.

SELF-CHECK

1. Given a 20-SMA of \$50 and a standard deviation of \$3, compute the upper and lower Bollinger Bands.
2. What does a Bollinger “squeeze” tell you — and what does it *not* tell you?
3. Compute the true range if today's high is \$30, low is \$27, and yesterday's close was \$25.
4. Why is ATR so useful for setting a stop-loss?

CHAPTER FIFTEEN

Using Indicators Wisely

Now you know how the main indicators are built. This short, vital chapter is about not fooling yourself with them — the discipline that separates a useful toolkit from a cluttered dashboard of self-deception.

IN THIS CHAPTER YOU WILL LEARN

- Why all indicators are derived from price (and what that implies).
- The danger of “indicator overload.”
- The trap of curve-fitting and optimisation.
- How to choose a small, complementary set.

Indicators are tools, and like any tool they can build or harm depending on the hand. The errors in this chapter are subtle, common, and quietly ruinous — and avoiding them matters more than knowing ten more indicators.

15.1 They All Come From Price

Here is a truth that reframes everything: **every indicator is just a mathematical transformation of price (and sometimes volume)**. RSI, MACD, moving averages, Bollinger Bands — they all take the same raw price data and re-package it. This means no indicator contains *new* information; it only makes some aspect of price easier to see. It also means indicators inevitably *lag* (they're computed from prices that already happened) and that stacking many of them gives a false sense of confirmation — they're all echoing the same price, like asking five translations of one sentence and mistaking the agreement for independent evidence.

15.2 Indicator Overload

Beginners, hungry for certainty, pile indicator upon indicator until the chart is a tangle of coloured lines — and then freeze, because the indicators inevitably contradict each other (one says buy, another sell). This **indicator overload** (or “analysis paralysis”) is one of the most common ways traders sabotage themselves. Since all indicators derive from price, ten of them tell you barely more than two well-chosen ones. Less is genuinely more. The seasoned trader's chart is usually *clean*.

15.3 The Curve-Fitting Trap

The most insidious error is **curve-fitting** (over-optimisation). It's tempting to tweak an indicator's settings until it would have produced perfect signals on *past* data — a 13.5-day RSI with a 68.3 threshold that nailed every turn last year. But a setting tuned to fit the past almost always fails on the future, because it was fitted to that period's *noise*, not to any real, repeating behaviour. The market's randomness guarantees that with enough tweaking you can “explain” any history; that explanation has zero predictive power. Beware any system, or any seller of systems, that looks flawless in backtest. Robust beats optimal: a simple setting that works *okay* across many conditions is worth more than a perfect one fitted to one stretch of history.

15.4 Choosing a Small, Complementary Set

The wise approach: pick a *few* indicators that measure *different* things, so each adds something the others don't. A sensible minimal kit might be one for trend (a moving average), one for momentum (RSI or MACD), and one for volatility (ATR or Bollinger Bands) — three tools answering three different questions: *which way, how strongly, and how wildly?* Adding a second momentum indicator to that set adds clutter, not insight. Combine indicators with the price-action skills of Parts I–III, keep the chart clean, and remember that the indicator is the servant of your judgement, never its replacement.

WATCH OUT**The seductive backtest**

Somewhere online, someone is selling an indicator or “system” with a backtest showing it turned \$1,000 into \$1,000,000. Assume it is curve-fitted, cherry-picked, or ignoring costs and slippage — because it almost always is. If such a system truly worked, its owner would quietly use it, not sell it to you for \$99. The existence of a perfect-looking backtest is, paradoxically, evidence *against* a system, not for it. Real edges are small, fragile, and hard-won — never the stuff of miracle equity curves.

KEY TAKEAWAYS

1. **All indicators derive from price** — they add no new information and inevitably lag.
2. **Indicator overload** causes contradiction and paralysis; a clean chart with a few tools beats many.
3. **Curve-fitting** (tuning settings to past data) produces flawless backtests and worthless predictions.
4. **Choose a small, complementary set** — trend, momentum, volatility — answering different questions.

SELF-CHECK

1. Why does stacking many indicators give a false sense of confirmation?
2. What is indicator overload, and why is a clean chart usually better?
3. Explain curve-fitting and why a perfect backtest is a red flag.
4. Name a sensible three-indicator kit and the question each one answers.

V

PART FIVE

Putting It Together

Tools are useless without a framework and a conscience. This final part turns technique into a disciplined approach, makes risk management the centre of everything, and ends with the honest truth about trading that almost no course will tell you.

AN APPROACH

| RISK MANAGEMENT

| THE HONEST TRUTH

CHAPTER SIXTEEN

Building a Trading Approach

A pile of patterns and indicators is not a plan. This chapter turns the tools of this book into a coherent, repeatable approach — and explains why having any consistent plan beats jumping between a hundred clever ideas.

IN THIS CHAPTER YOU WILL LEARN

- The elements of a written trading plan.
 - Why a defined edge and consistency matter more than any single signal.
 - The role of backtesting — and its honest limits.
 - Matching your approach to your temperament and time.
-

Most people who lose at trading don't lose for lack of a good signal; they lose for lack of a plan — acting on impulse, changing strategy after every loss, risking too much on a hunch. A written, tested approach is the antidote, and building one is what separates a trader from a gambler.

16.1 A Written Trading Plan

A trading plan is a written document answering, in advance, every decision you'll face — so you decide with a calm mind, not in the heat of a moving market. At minimum it specifies: **what** you trade (which markets/instruments), **your edge** (the specific, repeatable setup you wait for — e.g. “a bullish engulfing at a major support level on rising volume in an uptrend”), **entry** (exactly when you buy), **exit** (both your profit target *and* your stop-loss, decided before you enter), **position size** (how much you risk per trade — Chapter 17), and **your rules** (what you will and won't do). Writing it down is not bureaucracy; it's the only way to keep your future, emotional self honest.

16.2 Edge & Consistency

An **edge** is a setup that, over many trades, makes money *on average* — not every time, but with positive expectancy across the whole series. You don't need to be right often; you need your winners to outweigh your losers (Chapter 17's math shows how a trader right only 40% of the time can still profit handsomely). The key is **consistency**: applying the *same* tested edge again and again, so the law of averages can work in your favour. The trader who hops from strategy to strategy never gives any edge the sample size it needs to pay off, and never learns whether anything works.

16.3 Backtesting — and Its Limits

Before risking money, you can **backtest** a plan: apply its rules to historical data and see how it would have performed. Done honestly, this builds confidence and reveals flaws cheaply. But heed Chapter 15's warning about curve-fitting, and add these traps: a backtest that ignores *costs* (commissions, the bid-ask spread, slippage) flatters itself badly; one tested on too few trades proves nothing; and the past is not the future — a strategy that worked in a calm bull market may die in a crash. Backtesting tells you a plan *wasn't obviously broken in the past*; it cannot promise the future. Treat a good backtest as a green light to test small with real money, never as a guarantee.

16.4 Match the Approach to Yourself

There is no single “best” style; there's only what fits *you*. **Day trading** (in and out within a day) demands hours of screen time, fast reflexes, and an iron stomach — and has the worst odds of all (Chapter 18). **Swing trading** (holding days to weeks) suits those with a job and patience. **Position trading** and long-term **investing** (Books 1–4) demand the least time and have, by far, the best odds of building real wealth. Be brutally honest about your temperament, your free time, and your tolerance for stress and loss. The approach you can actually follow calmly beats the “optimal” one that makes you anxious and impulsive.

In Plain Terms · the plan is for the future you

When you write your plan, you are calm and rational. When the market is crashing or rocketing, you will *not* be — fear and greed hijack the clearest mind. The written plan is a letter from your rational self to your panicking future self, saying “we already decided this; follow the rules.” Discretionary, in-the-moment decisions are where most accounts go to die. The whole point of a plan is to remove yourself from the decision at the exact moment you're least fit to make it.

KEY TAKEAWAYS

1. **A written plan** defines what you trade, your edge, entry, exit (target + stop), size, and rules — in advance.
2. **An edge plus consistency** beats any single signal; you needn't be right often, just profitable on average.
3. **Backtest honestly** (include costs, enough trades) — it shows a plan wasn't broken, not that it'll work.
4. **Match the style to your temperament and time** — the plan you can follow calmly is the right one.

SELF-CHECK

1. List the core elements every trading plan should specify.
2. What is an “edge,” and why does consistency matter so much?
3. Give two ways a backtest can mislead you.
4. Why does the best trading style depend on the person, not the strategy alone?

CHAPTER SEVENTEEN

Risk Management — the Real Secret

If you remember only one chapter from this book, make it this one.

Professionals will tell you the truth that beginners refuse to hear: success in trading comes far more from managing risk than from predicting price. Here is the math that proves it.

IN THIS CHAPTER YOU WILL LEARN

- Why risk management — not prediction — is the real edge.
 - The stop-loss, and how to place it sensibly.
 - Position sizing: the 1% (or 2%) rule, worked with numbers.
 - Risk-reward ratio and why you can be wrong most of the time and still win.
-

Every chapter so far has hedged its tools with “this is only a probability.” This is the chapter that makes peace with that uncertainty — by showing that you don't need to predict well if you manage risk well. Survival first; profit follows.

17.1 Why Risk Management Is the Edge

Here is the counter-intuitive heart of trading: **you cannot control whether any trade wins, but you can completely control how much you lose when it doesn't.** A trader with mediocre signals but excellent risk control will outlast and outperform a brilliant forecaster who occasionally bets too big and blows up. The single fastest way to fail is the catastrophic loss — the one oversized position that wipes out months of gains. Risk management exists to make that impossible by design. Prediction is the part you can't control; risk is the part you can — so that's where the real edge lives.

17.2 The Stop-Loss

A **stop-loss** is a predefined price at which you will exit a losing trade, no questions asked — the seatbelt of trading. Set *before* you enter, it caps your loss and removes the most dangerous moment in trading: deciding whether to sell while watching money evaporate (when fear and hope will lie to you). Place it where your trade idea is *proven wrong* — e.g. just below the support level you bought at, or beyond normal volatility using ATR (Chapter 14), not at an arbitrary round number. And once set, **honour it**; the deadliest habit in trading is moving a stop-loss further away to avoid taking the loss, turning a small, planned loss into an account-destroying one.

17.3 Position Sizing — the 1% Rule (Worked)

Position sizing answers: *how many shares should I buy?* The professional's answer flows from a rule — risk only a small fixed fraction of your account on any single trade, commonly **1%** (conservative) or **2%**. This caps the damage of any one loss and makes a losing streak survivable. The formula ties together your account, your risk rule, and your stop-loss:

$$\begin{aligned}\text{Risk per trade} &= \text{Account} \times \text{risk \%} \\ \text{Position size (shares)} &= \text{Risk per trade} \div (\text{Entry price} - \text{Stop price})\end{aligned}$$

Let's work it fully. Account = **\$10,000**; risk rule = **1%**; you buy a stock at **\$50** and set your stop at **\$48** (a \$2 risk per share):

WORKED CALCULATION · POSITION SIZING (1% RULE)

Risk per trade = $10,000 \times 1\% = \textbf{\$100}$ (*the most you'll lose if stopped out*)

Risk per share = entry – stop = $50 - 48 = \textbf{\$2}$

Position size = $100 \div 2 = \textbf{50 shares}$

Capital deployed = $50 \times \$50 = \textbf{\$2,500}$

You buy 50 shares (\$2,500 of stock). If the stop hits at \$48, you lose $50 \times \$2 = \100 – exactly 1% of your account, as planned. Even ten losing trades in a row would cost only ~10%, leaving you very much alive to trade on.

Notice what this does: it makes your loss a *decision*, not an accident. The wider your stop, the *fewer* shares you buy — so risk per trade stays fixed regardless of the stock's volatility. This one discipline, applied without exception, is what keeps professionals in the game for decades.

17.4 Risk-Reward & Why You Can Be Wrong a Lot

The **risk-reward ratio** compares what you're risking to what you aim to gain. If you risk \$2 per share (entry \$50, stop \$48) to make \$6 (target \$56), that's a **1:3** risk-reward. Here's why this is liberating: with a 1:3 ratio you can be *wrong most of the time and still profit*. The math:

WORKED CALCULATION · WRONG 60% OF THE TIME, STILL WINNING

Setup: risk \$1 to make \$3 (1:3) on each trade, over 10 trades.

Win rate: only **40%** → 4 wins, 6 losses.

Wins: $4 \times (+\$3) = \mathbf{+\$12}$

Losses: $6 \times (-\$1) = \mathbf{-\$6}$

Net = $+12 - 6 = \mathbf{+\$6 \text{ profit}}$ – despite losing 60% of trades!

This is the whole game. Cut losses short, let winners run, keep risk-reward favourable – and you don't need to be a good forecaster at all. Most beginners do the exact opposite: small wins, huge losses, and a desperate need to “be right.”

WATCH OUT**The two account-killers**

Almost every blown-up trading account dies from one of two sins, both violations of this chapter. **One: moving the stop-loss** — refusing to take a small planned loss, letting it grow until it's catastrophic (the “it'll come back” lie). **Two: oversizing** — betting big on a “sure thing,” so a single loss does damage that dozens of wins can't repair. Neither has anything to do with chart-reading skill; both are failures of discipline. You can be the world's worst chart analyst and survive with good risk management — and the world's best and still die without it.

KEY TAKEAWAYS

1. **Risk management, not prediction, is the real edge** — you control your losses, not your wins.
2. **A stop-loss** set in advance (where your idea is proven wrong) caps the loss; never move it away.
3. **Position sizing (1% rule):** $\text{shares} = (\text{account} \times 1\%) \div (\text{entry} - \text{stop})$. (Worked: 50 shares, \$100 at risk.)
4. **With a 1:3 risk-reward you can be right 40% of the time and still profit** — cut losses, let winners run.

SELF-CHECK

1. Why is risk something you can control while prediction isn't?
2. Where should a stop-loss be placed, and what's the deadliest thing to do with one?
3. Account \$20,000, 2% rule, entry \$100, stop \$96 — how many shares, and what's the dollar risk?
4. Show, with numbers, how a 30% win rate can still be profitable at a 1:4 risk-reward.

CHAPTER EIGHTEEN

The Honest Truth About Trading

We end where most books won't go. You've learned the tools; now you deserve the truth about what they can realistically do for you — the truth that the industry selling courses, signals and brokerage commissions has every reason to hide.

IN THIS CHAPTER YOU WILL LEARN

- What the evidence really says about active trader returns.
- Why the odds are stacked against the short-term trader.
- The psychology that beats most people.
- How to use technical analysis sanely — and what most people should do instead.

This is the most important chapter in the book, and the one I'd most want a beginner to read. Everything before it was technique; this is wisdom. Read it twice.

18.1 What the Evidence Says

The research on active trading is remarkably consistent and remarkably grim. Across study after study, in market after market, **the large majority of active day traders lose money**, and only a tiny fraction are reliably profitable over the long run. Regulators in several countries now *require* brokers to display warnings that a majority of retail accounts lose money trading leveraged products — often 70–85% of them. This isn't pessimism; it's the documented base rate. The honest framing: short-term active trading is one of the hardest ways to make money that exists, and most who attempt it would have done better doing far less.

18.2 Why the Odds Are Stacked

The deck is stacked for structural reasons, not just skill. **Costs:** every trade pays a spread and often a commission; trade often and these compound into a serious drag — you must beat the market by enough to cover them just to break even. **Competition:** on the other side of your trade are professionals with faster data, lower costs, vast resources and, increasingly, algorithms reacting in microseconds. **Taxes:** short-term gains are often taxed more heavily than long-term holdings. **Randomness:** as Chapter 1 warned, much of what looks tradeable is noise. And **leverage** (trading with borrowed money), heavily marketed, magnifies losses as much as gains and is the single fastest route to ruin. The amateur trades against all of this at once.

18.3 The Psychology That Beats Most People

Even with good tools and good risk rules, the final boss is your own mind. **Fear** makes you cut winners too early and freeze when you should act. **Greed** makes you oversize, chase, and ignore your plan. **Hope** makes you hold losers, moving the stop you swore you'd honour. **Revenge trading** — trying to win back a loss immediately — turns one bad trade into ten. The market is an engine perfectly designed to exploit these instincts, and self-control under real financial pressure is far rarer and harder than any chart skill. Most people who fail at trading don't fail at analysis; they fail at managing themselves. This is why the disciplines of Chapters 16–17 matter more than every pattern in Parts II–III combined.

18.4 How to Use This Book Sanely

So what should you actually *do* with everything you've learned? An honest set of conclusions:

- **For most people, the path to wealth is not trading at all** — it's the patient, diversified, long-term investing of Books 1–4: buy good assets or low-cost index funds, keep costs and taxes low, and let compounding work over decades. The evidence overwhelmingly favours this over active trading.
- **Technical analysis is still genuinely useful** — as a way to understand market behaviour, to time entries and exits a little better, to impose discipline, and (above all) to manage risk. Used as a *supplement* to sound investing, with humility, it earns its place.

- **If you do choose to trade, do it honestly:** start with money you can truly afford to lose, paper-trade or trade tiny first, treat it partly as paid education, obey your risk rules religiously, and measure yourself against simply having bought an index fund. Many who do the honest comparison quietly go back to indexing — and are richer for it.
- **Be deeply sceptical of anyone selling certainty** — guru signals, miracle systems, “secret” patterns, get-rich-quick courses. The people who consistently make money from retail trading are mostly the ones *selling to* traders, not trading themselves.

THE BOTTOM LINE

A craft worth knowing, a humility worth keeping

Technical analysis is a real craft — a useful language for reading market mood, timing, and risk. It is not a money machine, a crystal ball, or a substitute for patient investing, and anyone who tells you otherwise is selling something. Learn it for what it honestly is: a set of probabilistic tools and, most of all, a framework for managing risk. Hold it with humility, pair it with the long-term wisdom of the earlier books, protect your capital above all — and you’ll be far ahead of the crowd who came chasing certainty and left poorer. That humility, more than any pattern or indicator, is the real lesson of this book.

KEY TAKEAWAYS

1. **The evidence is clear:** most active/day traders lose money; only a small minority profit over the long run.
2. **The odds are stacked** by costs, professional competition, taxes, randomness and (worst) leverage.
3. **Psychology — fear, greed, hope, revenge — beats most people**, which is why discipline outranks analysis.
4. **For most, patient long-term investing wins;** use technical analysis as a humble supplement and protect capital first.

SELF-CHECK

1. What does the broad evidence say about the profitability of active day trading?
2. Name three structural reasons the odds are stacked against short-term traders.
3. Which four emotions most often sabotage traders, and how?
4. What is the sanest way for most people to build wealth, and what role should technical analysis play?

REFERENCE

Glossary

The working vocabulary of technical analysis, in plain language.

ATR (Average True Range) — a volatility indicator: the average size of a period's price range (counting gaps). Used to size stop-losses.

Bollinger Bands — a channel of a 20-period SMA $\pm$ 2 standard deviations; widens with volatility, pinches in a “squeeze.”

Breakout — when price moves decisively through a support or resistance level; more trustworthy on heavy volume.

Candlestick — a chart symbol showing one period's open, high, low and close; body = open-to-close, wicks = the extremes.

Continuation pattern — a consolidation (triangle, flag, pennant) after which the prior trend usually resumes.

Curve-fitting — over-tuning a strategy's settings to past data; produces great backtests and poor real results.

Divergence — when an indicator disagrees with price (e.g. price higher high, RSI lower high); an early warning of fading momentum.

Doji — a candle with almost no body (open $\approx$ close); signals indecision.

EMA (Exponential Moving Average) — a moving average weighting recent prices more heavily; multiplier $k = 2/(N+1)$.

Engulfing — a two-candle reversal where a large candle's body swallows the previous one.

Gap — an empty space where price leaps from one close to the next open with no trades between; usually news-driven.

Golden / death cross — when the 50-day SMA crosses above (golden) or below (death) the 200-day SMA.

Hammer / hanging man — same shape (small body, long lower wick): bullish after a fall, bearish after a rise.

Head and shoulders — a reversal pattern of three peaks (middle highest); confirmed on a break of the neckline.

MACD — momentum indicator: $EMA_{12} - EMA_{26}$, with a 9-EMA signal line and a histogram of their difference.

Marubozu — a long candle with no wicks; one side controlled the whole period.

Moving average — a smoothed line of the last N prices, revealing the trend; inherently lags.

Overbought / oversold — RSI above 70 / below 30; indicates stretched momentum, *not* an automatic sell/buy.

Position sizing — deciding how many shares to buy so that a loss equals a fixed small % of the account.

Resistance — a price “ceiling” where selling repeatedly halts a rise.

Risk-reward ratio — the amount risked versus the amount targeted (e.g. 1:3); favourable ratios let you profit while often wrong.

RSI (Relative Strength Index) — momentum oscillator (0–100): $100 - 100/(1+RS)$, $RS = \text{avg gain} / \text{avg loss}$.

SMA (Simple Moving Average) — the plain average of the last N closing prices.

Stop-loss — a predetermined exit price that caps a trade's loss; should never be moved further away.

Support — a price “floor” where buying repeatedly halts a decline.

Trend — the prevailing direction: uptrend (higher highs & lows), downtrend (lower highs & lows), or sideways.

Trendline — a straight line connecting rising lows (uptrend) or falling highs (downtrend).

Volume — the number of shares traded in a period; measures conviction and should confirm a move.

Whipsaw — a false signal in a choppy market that reverses immediately, causing repeated small losses.

REFERENCE

Sources & Further Reading

Where to verify the formulas, the conventions, and the sobering evidence — and where to go deeper.

• Foundational texts

Murphy, John J. — *Technical Analysis of the Financial Markets*. The standard comprehensive reference; covers trend, patterns and indicators in depth.

Nison, Steve — *Japanese Candlestick Charting Techniques*. The book that introduced candlesticks to Western traders; definitive on the patterns of Part II.

Edwards & Magee — *Technical Analysis of Stock Trends*. The classic origin of chart-pattern analysis (Part III).

Wilder, J. Welles — *New Concepts in Technical Trading Systems*. The original source for the RSI and ATR formulas (Chapters 13–14).

Bollinger, John — *Bollinger on Bollinger Bands*. The creator's own account (Chapter 14).

• On the honest odds (Chapter 18)

Barber & Odean — “Trading Is Hazardous to Your Wealth” and related studies, showing active individual traders underperform.

Barber, Lee, Liu & Odean — studies of Taiwanese day traders finding the large majority lose money over time.

Regulatory disclosures — under rules such as the EU's ESMA, brokers must publish the percentage of retail accounts that lose money on leveraged products (commonly 70–85%); check any broker's own figure.

Malkiel, Burton — *A Random Walk Down Wall Street*. The classic case for low-cost index investing over active trading.

• Verifying the math

Every formula in Part IV — SMA, EMA ($k = 2/(N+1)$), RSI, MACD, Bollinger Bands, ATR — is standard and can be cross-checked against the original sources above or any reputable financial-education reference. The worked numbers in this book were computed by hand; redo them yourself as the self-checks ask, and confirm they match. Understanding beats trusting.

A note on data: to practise, pull free historical price data from your brokerage or public sources and recompute these indicators on a spreadsheet. Seeing an indicator built from raw prices — exactly as Chapter 12 did by hand — is the fastest way to truly understand (and stop over-trusting) it.

WHERE THIS SITS IN THE SERIES

This is **Book 5** of *Money, Mastered — The Roadmap*. It builds on the foundations of **Book 1 (Money, From Zero)** and **Book 2 (Where to Put Your Money)**, complements the business-analysis craft of **Book 4 (Fundamental Analysis)**, and — as Chapter 18 stresses — should always be read alongside, never instead of, the patient long-term investing those books teach. Next in the roadmap: funds, SIPs and portfolios; then derivatives, tax, psychology and safety; then building and protecting wealth.
